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MASTER THESIS

Handling Missing Data in Multilevel Models with Few Clusters: A Simulation Study

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Abstract

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by Lona Frießner

Multilevel studies often involve two challenges simultaneously: small numbers of clusters and missing data. However, the performance of modern missing-data methods under these combined conditions remains insufficiently understood, particularly in models with missing predictor variables and contextual effects. This thesis examines the performance of multiple imputation, using both conventional and small-sample-corrected degrees of freedom, and compares it with a fully Bayesian approach in a simulation study. The study features a simple random-intercept model with a manifest group-mean predictor, with missings in the predictor variable and one group-level auxiliary variable. Simulation conditions varied in terms of intraclass correlation (0.1 vs. 0.3), missingness mechanism (MAR vs. MCAR), between-group effect size (0 vs. .4), and level-2 sample size (15, 30, 60). Method performance was evaluated in terms of bias, coverage, standard deviation of point estimates, and interval width. Results indicate a trade-off between bias and interval width for the between-group effect: multiple imputation tended to produce more biased estimates, particularly under low intraclass correlation conditions, whereas the Bayesian approach yielded wider credible intervals. As coverage under conventional degrees of freedom was already close to the nominal level, the small-sample correction tended to result in overcoverage. The findings are discussed with regard to methodological limitations and their implications for applied multilevel research.

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Chapter 1

Introduction

Multilevel models are a standard statistical tool for analyzing clustered data, which frequently occur in the social sciences (e.g., patients nested within therapists, students within classes, or employees within firms). In practice, such data sets are often characterized by a relatively small numbers of groups and by the presence of missing values.

Methodological research has proposed a range of approaches to address each of these challenges separately. Small-sample corrections for multilevel models have been developed to improve statistical inference when the number of clusters is limited (Elff et al., 2021; Holtmann et al., 2016; McNeish & Stapleton, 2016b, 2016a). At the same time, modern missing-data methods have been extended to multilevel data structures (see e.g. Audigier et al., 2018; Enders et al., 2016; Enders, 2025; Grund et al., 2019; Kahan et al., 2016; Yucel, 2008).

Some research has investigated how these two challenges interact, both for single-level regressions (see e.g. McNeish, 2017a; Von Hippel, 2013) and in multilevel contexts (e.g., McNeish, 2018; Shin et al., 2025; Shin & Shin, 2025; Zitzmann et al., 2015). However, the existing literature remains limited, and many relevant scenarios have not yet been systematically investigated. Among these, situations in which predictor variables contain missing values and are simultaneously used to construct group-mean contextual effects represent one particularly important but understudied case.

In applied research, effects at both the individual and the group level are often of interest and can be modeled within multilevel frameworks. Level-2 characteristics are frequently operationalized by aggregating Level-1 variables, resulting in so-called contextual effects (Croon & Van Veldhoven, 2007; Lüdtke et al., 2008; Snijders & Bosker, 2012, p. 102). This approach is widely used in psychological research (e.g., Anderman, 2002; Papaioannou et al., 2004). At the same time, missing values in predictor variables pose additional challenges for statistical inference, as they complicate missing data handling compared to situations in which only the outcome variable is incomplete. Standard likelihood-based approaches such as full information maximum likelihood are not easily applicable in this setting.

This thesis therefore investigates the suitability of modern approaches to handling missing data in small-sample multilevel models in which predictor variables are partially missing and simultaneously included as group-mean contextual predictors. Specifically, the performance of multiple imputation and fully Bayesian estimation is evaluated.

The structure of this thesis is as follows: First, the theoretical foundations of multilevel models are introduced, and current methodological approaches for handling small samples and missing data are reviewed. Subsequently, a simulation study is conducted to investigate the performance of these methods under different data conditions.

Chapter 2

Theory

2.1 Theoretical foundation of multilevel models

Example

Imagine you are interested in the musical self-concept of young adults and whether it is related to how many hours they practice their instrument each week. To investigate this question, you collect data from several youth ensembles. You recruit 15 ensembles, each consisting of 10 players. Every player completes a questionnaire assessing their musical self-concept as well as their weekly practice time.

At first glance, this seems like a simple data set of 150 individuals. But is it really that simple?

Players from the same ensemble rehearse together, share teachers, and experience similar social and musical environments. Their responses may therefore be more similar within ensembles than between ensembles.

Populations like this are common in the social sciences. Units of interest (e.g., students, employees, musicians) are grouped into higher-level units such as classes, firms, or ensembles. The resulting data is called *nested* or *clustered* (Snijders & Bosker, 2012, p. 7). From a sampling perspective, such data often arise from a multistage sampling process, in which groups are sampled first and individuals are subsequently sampled within these groups¹. This process induces dependencies in the data, as observations within the same group are not selected independently. Beyond the sampling process, individuals within the same group typically share contextual influences (e.g., the same ensemble teacher or classroom environment), which leads to increased similarity in their responses. As a consequence, the assumption of independent errors in traditional linear models is violated, because residuals within groups are correlated (Raudenbush & Bryk, 2010, p. xx). A widely used approach to account for this structure is the use of *multilevel models* (also referred to as *hierarchical linear models* or *mixed models*), which are introduced in the following section.

2.1.1 The concept of multilevel regression models

The aim of regression analysis, whether single- or multilevel, is to explain variance in an outcome variable by including associated variables in the model as predictors

¹In this thesis, the terms ‘groups’ or ‘clusters’ refer to higher-level units, while lower-level units are referred to as ‘persons’, ‘individuals’ or ‘observations’. These models can also be applied to longitudinal data, where measurement occasions are nested within individuals, and to data structures with more than two levels.

(Snijders & Bosker, 2012, p. 80). In single-level regression, the residual e_i captures the unexplained variance in Y , that is differences in the outcome from the prediction that the model made (Leonhart, 2013, p. 319). The assumption of the classical linear model is that after inclusion of all systematic influences on Y , the residuals are random. Specifically, they are assumed to have expectation zero ($E(e_i) = 0$), constant variance across observations (homoscedasticity; $Var(e_i) = \sigma^2$), and to be independent across observations ($Cov(e_i, e_{i'}) = 0$ for $i \neq i'$) (Fahrmeir et al., 2021, p. 87).

Multilevel models extend regular linear regression by adding level-2 random effects to explain the variability between groups that cannot be accounted for by predictors (Snijders & Bosker, 2012, p. 80). This partition of variance can be expressed in the multilevel regression model without predictors, the *empty model*:

$$Y_{ij} = \gamma_{00} + u_{0j} + e_{ij}$$

Here, the outcome value of person i in group j is modeled as the sum of the overall mean γ_{00} , a group level random effect u_{0j} and an individual-level residual, e_{ij} (Snijders & Bosker, 2012, p. 49). The random intercept u_{0j} captures the dependency of individuals within one group statistically, as all individuals in group j have the same u_{0j} value and their outcome values are therefore correlated. Conditional on this group component, however, the remaining level-1 residuals e_{ij} are assumed to be independent, addressing the issue of dependency that would arise if a single-level regression model were applied to nested data.

This and the models explained below are illustrated in Figure 2.1.

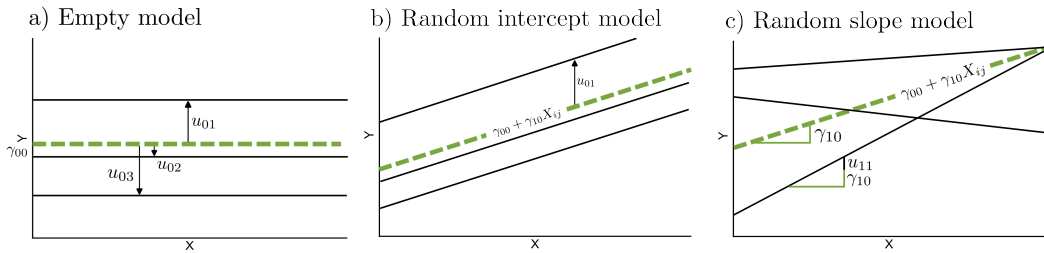


FIGURE 2.1: Schematic illustration of common multilevel model specifications in coordinate space: empty model, random intercept model, and random slope model. The green line represents the fixed-effect prediction averaged across groups. The black lines represent exemplary groups and their deviations from the fixed effects. Individual-level observations are not displayed.

2.1.2 Within- and between- group regressions and inclusion of predictors

Predictors can then be included to explain variability within groups (level 1) and between groups (level 2).

At level 1, the outcome of a person i in group j is modeled as a function of individual-level explanatory variables X_{ij} to obtain the within-group regression model. For simplicity, the model is illustrated with a single level-1 predictor:

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + e_{ij} \quad (2.1)$$

At level 2, the variation between groups in the group-dependent coefficients β_{0j} and β_{1j} is modeled. In a random coefficients model, each group is allowed to have its own intercept and slope. These coefficients can be separated into a fixed part (the average coefficient across groups) and a random deviation from this average (Snijders & Bosker, 2012, p. 75). Without level-2 predictors:

$$\beta_{0j} = \underbrace{\gamma_{00}}_{\text{Fixed part}} + \underbrace{u_{0j}}_{\text{Random part}}$$

$$\beta_{1j} = \underbrace{\gamma_{10}}_{\text{Fixed part}} + \underbrace{u_{1j}}_{\text{Random part}}$$

On this level, in order to explain between-group variability in the intercepts and slopes, level-2 predictors Z_j may be included (Snijders & Bosker, 2012, p. 80). Including such predictors may reduce the unexplained variance of the random effects u_{0j} and u_{1j} . With a single level-2 predictor, the model becomes:

$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j} \quad (2.2)$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j} \quad (2.3)$$

Substituting the level-2 equations Equation 2.2 and Equation 2.3 into the level-1 model Equation 2.1 yields the combined multilevel model:

$$Y_{ij} = \underbrace{\gamma_{00} + \gamma_{10}X_{ij} + \gamma_{01}Z_j + \gamma_{11}X_{ij}Z_j}_{\text{Fixed part}} + \underbrace{u_{0j} + u_{1j}X_{ij} + e_{ij}}_{\text{Random part}}$$

Y_{ij} : Outcome for individual i in group j

X_{ij} : Level-1 (within-group) predictor

Z_j : Level-2 (between-group) predictor

γ_{00} : Fixed intercept (overall mean when $X_{ij} = 0$, $Z_j = 0$)

γ_{10} : Average within-group effect of X_{ij} on Y_{ij}

γ_{01} : Effect of Z_j on the intercept (between-group effect)

γ_{11} : Cross-level interaction; change in the effect of X_{ij} depending on Z_j

u_{0j} : Random intercept (group-specific deviation from γ_{00})

u_{1j} : Random slope (group-specific deviation from γ_{10})

e_{ij} : Level-1 residual

If no level-2 predictors are included, the model reduces to a random-coefficients model without contextual predictors.

2.1.3 Distributional assumptions of the multilevel regression model

Similar distributional assumptions are made for the level-2 random effects as for the residuals in single-level regression: they have mean zero, constant variance across clusters, and are independent of each other across clusters and independent of the level-1 residuals. However, random effects within the same cluster - i.e. the random intercept u_{0j} and random slope u_{1j} - may be correlated and are often expected to do so (Snijders & Bosker, 2012, p. 76). Additionally, residuals and random effects are commonly assumed to follow a normal distribution for convenience, as it facilitates

the likelihood-based estimation of the unknown parameters (Fahrmeir et al., 2021, p. 388). In summary:

$$\begin{aligned} \begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} &\sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{T} \right), & \mathbf{T} &= \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix} \\ e_{ij} &\sim \mathcal{N}(0, \sigma^2) \\ \text{Cov}(u_{0j}, e_{ij}) &= 0, & \text{Cov}(u_{1j}, e_{ij}) &= 0 \end{aligned}$$

2.1.4 The random intercept model

In many empirical applications of multilevel models the slope β_{1j} is assumed to be constant across groups (i.e., no random slope component, $u_{1j} = 0$), leading to the *random intercept model* (Snijders & Bosker, 2012, p. 45). This model allows for baseline differences in the outcome between groups, whereas the strength of the relationship between predictors and the outcome is assumed to be constant across groups:

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + \gamma_{01}Z_j + u_{0j} + e_{ij}$$

The random intercept model also provides a convenient framework for examining the use of group means of a given level-1 predictor as additional level-2 predictors. Including the group mean as a contextual variable allows the within-group regression coefficient to vary from the between-group regression coefficient (Snijders & Bosker, 2012, p. 57). The following example illustrates how processes operating within ensembles may differ from those operating between ensembles:

Example

Students often compare their own practice behavior with that of other ensemble members. A student who practices more than their peers may feel more competent and report a higher musical self-concept. At the same time, students in ensembles where everyone practices a lot may feel less competent because upward social comparisons become more likely.

This leads to two possible hypotheses:

- Students who practice more than others in their ensemble tend to report a higher musical self-concept.
- Students in ensembles with a higher average practice time may report a lower musical self-concept.

Including this separation allows for the analysis of so-called *contextual effects* that are frequently observed in empirical settings (Snijders & Bosker, 2012, p. 27). In such cases, the aggregate of the person-level characteristic has an effect on the outcome variable even after controlling for the effect of the individual characteristic (Raudenbush & Bryk, 2010, p. 139). To separate within- and between-group effects, the level-1 predictor is often group-mean-centered in such analyses: Instead of using X_{ij} directly, the centered predictor $(X_{ij} - \bar{X}_j)$ is entered into the model. This decomposes within- and between- group effects and facilitates their interpretation (Snijders & Bosker, 2012, p. 58). A random intercept model including such a decomposition can be written like

this²:

$$Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{.j}) + \gamma_{01}\bar{X}_{.j} + u_{0j} + e_{ij} \quad (2.4)$$

In this random intercept model with a contextual effect, the musical self-concept of student i in ensemble j is predicted by the sum of several components:

- γ_{00} : the average musical self-concept across all ensembles
- $\gamma_{10}(X_{ij} - \bar{X}_{.j})$: the effect of the student's practice time relative to the average practice time in their ensemble
- $\gamma_{01}\bar{X}_{.j}$: the effect of the ensemble's average practice time
- u_{0j} : a random deviation of ensemble j 's average musical self-concept from the overall mean (unexplained differences between ensembles)
- e_{ij} : a random deviation of the student's musical self-concept from the predicted value within their ensemble

The regression coefficients represent two conceptually different effects:

- γ_{10} : the *within-group effect*, indicating how a student's own practice time relates to their musical self-concept while controlling for the ensemble context
- γ_{01} : the *between-group effect*, indicating how the ensemble's average practice time relates to students' musical self-concept, controlling for individual deviations from the group mean

The difference between these two coefficients, $\gamma_{01} - \gamma_{10}$, represents the *contextual effect*. For the remainder of this thesis, the focus lies on this particular random intercept model.

2.1.5 The intraclass correlation coefficient (ICC)

In the random intercept model, the variance components are given by

$$u_{0j} \sim \mathcal{N}(0, \tau_0^2), \quad e_{ij} \sim \mathcal{N}(0, \sigma^2), \quad \text{Cov}(e_{ij}, u_{0j}) = 0.$$

From these variance components, the proportion of variability that is attributable to between-group differences can be calculated. This quantity is referred to as the *intraclass correlation coefficient* (ICC). In the empty model, the ICC is equal to the correlation between the outcomes of two randomly chosen individuals from the same group (Snijders & Bosker, 2012, p. 18). In models including predictors, the ICC is interpreted conditionally on the predictors, that is, as the correlation between two individuals from the same group who share identical predictor values (also referred to as the *residual ICC*) (Snijders & Bosker, 2012, p. 52).

Formally, the ICC is defined as

$$ICC = \frac{\tau_0^2}{\tau_0^2 + \sigma^2}$$

The ICC is an important diagnostic metric because it indicates whether the use of a multilevel model is advantageous. If $ICC = 0$, no between-group variance is present and a conventional single-level regression model is often sufficient. Conversely, larger ICC values indicate increasing similarity within groups and thus statistical dependency

²The group mean $\bar{X}_{.j}$ is treated as an observed (manifest) aggregate in this approach. Group means can also be modeled as latent variables, thereby accounting for measurement error and sampling variability (Lüdtke et al., 2008)

of observations. Ignoring this dependency and applying a classical linear model leads to biased standard errors, confidence intervals, and hypothesis tests (Fahrmeir et al., 2021, p. 370).

2.2 Estimation of multilevel models

In the preceding chapter, the theoretical foundations and parameters of the multilevel model were introduced. In practice, however, these parameters are unknown and only exist in the theoretical realm as ‘true’ values. The fixed effects γ and variance components σ^2 and τ^2 must therefore be estimated from observed data. Based on these estimates, inferences about the underlying population parameters are drawn. This process is illustrated for the random intercept model in Figure 2.2. Provided that the model specification corresponds to the true data-generating process well enough, the reliable performance of this ‘inferential machinery’ depends on three main elements (Elff et al., 2021):

1. Estimation of fixed-effect coefficients (regression parameters)
2. Estimation of variance–covariance components of the random effects (u) and residual errors (e)
3. Selection of an appropriate approximation of the sampling distribution of the test statistic.

Aspects of the observed data may pose challenges to this estimation process. Moreover, the inferential machinery itself (i.e., its formulas and underlying assumptions) may not adequately reflect the properties of the observed data. Two common challenging aspects, namely small sample sizes and missing data, will be discussed in the following chapters.

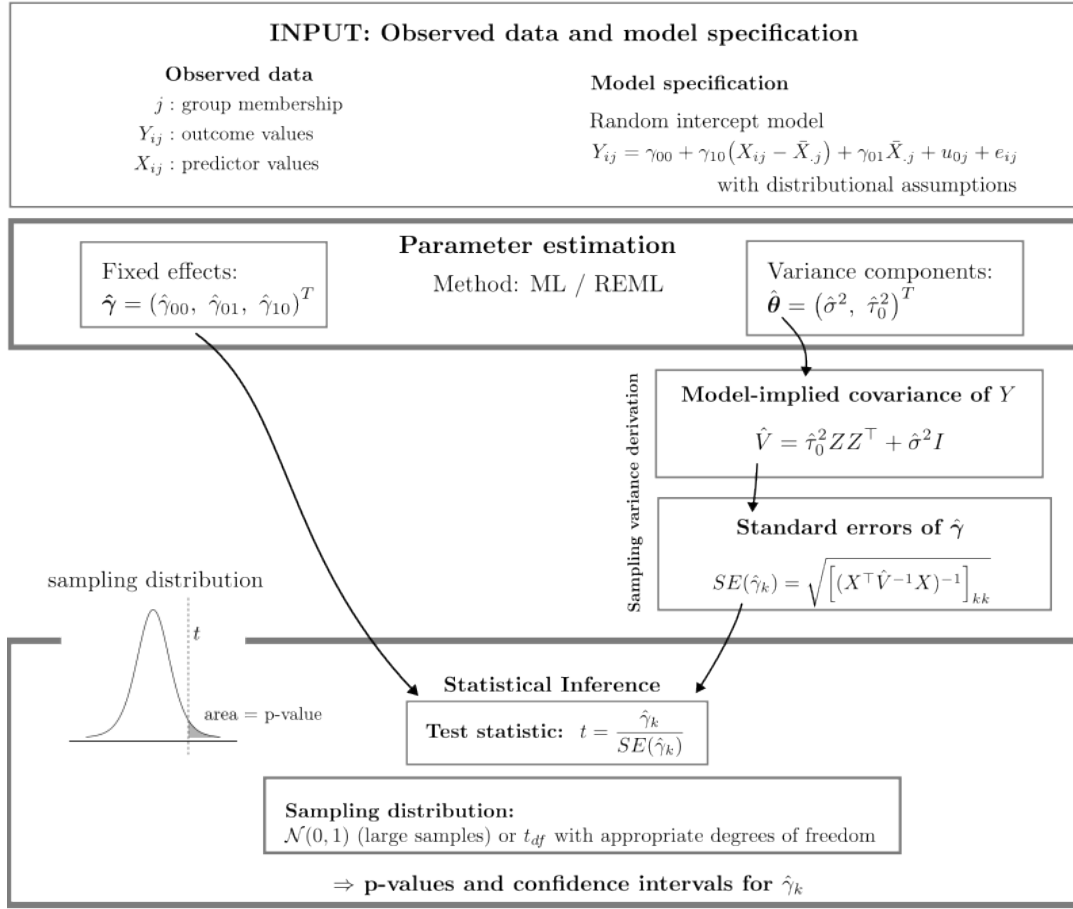
2.3 Small sample sizes in multilevel models

Example

Consider again the musical self-concept study with 150 students nested in 15 ensembles. Although 150 participants may seem like a reasonably large sample, the number of ensembles is still small. Information about between-ensemble variation and level-2 effects is therefore based on only 15 higher-level units. For this reason, such a design would typically be considered a small-sample multilevel study. What are the implications for statistical inference in such a study?

Frequentist estimation procedures for linear multilevel models typically rely on large-sample (asymptotic) theory. In multilevel models, the number of groups is particularly important. Simulation studies suggest that approximately 25–30 clusters are often required for reliable estimation, depending on the model specification and the parameters of interest (McNeish, 2017a; McNeish & Stapleton, 2016b). In practice, however, such numbers are often difficult to achieve because recruiting a large number of groups can be challenging and costly.

When the number of groups is small, several components of the inferential procedure may perform poorly, particularly the estimation of variance components, the calculation of standard errors, and the approximation of sampling distributions. The main issues associated with small-sample multilevel models and available corrections and alternatives are presented below.



Illustrative example. For a hypothetical data set with four persons nested in two groups:

The corresponding model matrices are

$$X = \begin{pmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 3 \\ 1 & 5 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \quad Y = \begin{pmatrix} 5 \\ 4 \\ 2 \\ 5 \end{pmatrix}$$

Illustrative observed values

id	j	X_{ij}	Y_{ij}
1	A	2	5
2	A	1	4
3	B	3	2
4	B	5	5

and the implied covariance matrix is

$$\hat{V} = \begin{pmatrix} \hat{\tau}_0^2 + \hat{\sigma}^2 & \hat{\tau}_0^2 & 0 & 0 \\ \hat{\tau}_0^2 & \hat{\tau}_0^2 + \hat{\sigma}^2 & 0 & 0 \\ 0 & 0 & \hat{\tau}_0^2 + \hat{\sigma}^2 & \hat{\tau}_0^2 \\ 0 & 0 & \hat{\tau}_0^2 & \hat{\tau}_0^2 + \hat{\sigma}^2 \end{pmatrix}$$

Schematic representation of the frequentist inferential process in a random intercept model, based on Snijders & Bosker (2012) and Raudenbush & Bryk (2010).

Notes on notation

- X : design matrix for the fixed effects. Each row corresponds to an observation (e.g., a person) and each column to a predictor included in the fixed part of the model.
- Z : design matrix for the random effects. In the random intercept model it encodes group membership (1 = observation belongs to the group, 0 = otherwise), so observations from the same group share the same random intercept.
- Y : outcome vector containing the observed values of the dependent variable.
- $\hat{V}$: estimated variance-covariance matrix of the outcome vector Y . In the random intercept model it is determined by the estimated variance components of the random intercept ($\hat{\tau}_0^2$) and the residual variance ($\hat{\sigma}^2$), reflecting that observations within the same group are correlated.
- $\hat{\gamma}_k$: the k -th element of the estimated fixed-effect vector $\hat{\gamma}$.

FIGURE 2.2

2.3.0.1 Estimation of regression parameters

The regression parameters are estimates of the effects of interest. For example, how much does one additional hour of weekly practice influence a student's musical self-concept? How does the musical self-concept differ between ensembles that practice on average three hours per week and those that practice thirteen hours per week?

Kackar & Harville (1981) showed that both ML and REML estimators of variance components are even and translation-invariant, implying that fixed effect point estimates are unbiased even in small samples.

2.3.0.2 Estimation of variance components

The estimation of the variance components, which are directly involved in the estimation of standard errors of the fixed effects, is more delicate. In ML estimation, fixed effects and variance components are estimated simultaneously, typically via iterative procedures like the Expectation Maximization (EM) mechanism (McNeish, 2017b). In this process fixed effects are effectively being treated as known when estimating variance components. Consequently, the sampling variability of the fixed-effect estimates is ignored and the loss of degrees of freedom due to their estimation is not accounted for. While these issues are not troublesome in large samples, they lead to systematic underestimation of variance components in small samples (McNeish, 2017b).

REML addresses this issue by separating the estimation conceptually. The fixed effects are eliminated from the likelihood, and variance components are estimated solely based on residual information. Fixed effects are then estimated via generalized least squares (GLS), using the previously estimated variance components (Raudenbush & Bryk, 2010, p. 410). This leads to less downward-biased variance component estimates, while ML and REML become asymptotically equivalent as sample size increases. Nevertheless, the estimated variance of the fixed-effect estimators may still be too small in small samples for two reasons. First, the variance formula treats the estimated variance components $\hat{\theta}$ as known and therefore ignores their sampling variability (Kackar & Harville, 1984). Second, the REML variance expression itself relies on asymptotic approximations (McNeish, 2017b). Both issues can lead to underestimated standard errors. Several small-sample corrections have been proposed to address this problem, among which the Kenward–Roger procedure is one of the most widely used (McNeish, 2017b).

2.3.0.3 Approximation of the sampling distribution

The standardized test statistic used for inference (see Figure 2.2) can be regarded as 'small-sample-adjusted' at this point. The final step concerns the choice of a sampling distribution for p-values and confidence intervals. In small samples, a t -distribution is typically recommended instead of a standard normal distribution (Snijders & Bosker, 2012, p. 95). Compared to the standard normal distribution, the t -distribution has heavier tails, particularly when the degrees of freedom are small. As a result, it yields wider confidence intervals and larger p-values, making statistical inference more conservative in small samples.

The shape of the t -distribution is determined by the number of degrees of freedom. With small degrees of freedom, even small changes noticeably affect the shape of the sampling distribution and therefore the resulting p-values and confidence intervals. The concept of degrees of freedom is difficult to define concisely, but it can be described as the number of independent pieces of information in the data that remain after accounting for restrictions such as estimated parameters (Pandey & Bright, 2008).

In multilevel models, the degrees of freedom generally cannot be calculated exactly and must therefore be approximated because of the nested data structure (Snijders & Bosker, 2012, p. 95). The Kenward–Roger correction mentioned earlier also includes a complex small-sample-adjusted approximation of the degrees of freedom, building on the Satterthwaite method (McNeish, 2017b).

A more straightforward approximation is provided in textbooks by Raudenbush & Bryk (Raudenbush & Bryk, 2010, p. 58) and Snijders & Bosker (Snijders & Bosker, 2012, p. 95), and has also been proposed in methodological articles (Elff et al., 2021). This approach distinguishes between level-1 and level-2 predictors and reflects the concept of degrees of freedom as information minus estimated parameters:

$$df_{Lv1} = N_1 - r - 1, \quad df_{Lv2} = N_2 - q - 1 \quad (2.5)$$

N_1 : Level-1 sample size (total number of individuals)

N_2 : Level-2 sample size (number of groups)

r : Total number of predictors

q : Number of predictors at level two

2.3.1 Fully Bayesian analysis

As alternative to the frequentist framework predominantly presented here, a Bayesian approach is often recommended for small sample sizes (McNeish, 2016a). In this framework, parameters are viewed as random variables with probability distributions. Prior knowledge about the parameters of interest in the form of prior distributions is combined with the information in the observed data in the form of a likelihood function. Bayes' theorem yields the posterior distribution, representing the updated knowledge about the parameters (Van De Schoot et al., 2014). More general details on Bayesian analyses will not be discussed in this thesis. Two aspects of Bayesian analyses are particularly helpful in small samples.

First, Bayesian inference does not rely on large-sample asymptotic approximations. Instead, the posterior distribution, the end result of a Bayesian analysis, is typically approximated using Markov Chain Monte Carlo (MCMC) sampling. Consequently, Bayesian inference does not require small-sample corrections for standard errors or degrees of freedom.

Second, Bayesian methods allow the incorporation of prior information about parameters. Informative or weakly informative priors may improve estimation accuracy by adding information when the likelihood is relatively flat in small samples (Gelman et al., 2025, p. 355; Smid et al., 2020). In addition, prior distributions may prevent inadmissible parameter estimates that could otherwise lead to convergence problems, such as negative variance estimates (Van De Schoot et al., 2014). At the same time, prior specification becomes particularly influential under such conditions. Vague or software default priors may therefore have unintended effects and can perform poorly in small samples (Gelman, 2006; McNeish, 2016a, 2016b). Thus, the inclusion of prior distributions may be regarded either as a methodological advantage or as “a price to be paid” (Raudenbush & Bryk, 2010, p. 411) for Bayesian inference.

2.4 Missing data in multilevel models

Example

In the study about musical self-concept, you find after data collection that some students in several ensembles did not report how many hours they practice per week. They completed the questionnaire on musical self-concept but left the practice question blank.

The reason for this is unknown: they might have forgotten to answer the question, skipped it intentionally, or misunderstood it. Regardless of the reason, the data set now contains missing values. How should these missing observations be handled in the analysis?

Missing data (i.e., incomplete data sets) are a widespread issue in empirical research and of course also apply to multilevel data, e.g., because individuals drop out of studies or forget to fill out certain items of a questionnaire. Three major concerns commonly arise from data that contain missing values (Lüdtke et al., 2007):

1. The effective sample size is reduced, leading to lower efficiency in estimation (this is especially troublesome if the would-be sample size is already small)
2. Typical statistical methods are difficult to conduct, because they were designed for complete data.
3. Parameter estimation may be biased, if the missing and the observed data differ from each other systematically

Therefore, it is crucial to treat missing data appropriately. In the last three decades, this issue has received attention in psychological research, and modern methods were developed (Enders, 2022; Schafer & Graham, 2002), more recently also for the more complex case of multilevel data (Enders, 2025; Lüdtke et al., 2017). This chapter discusses these methods, after giving a short introduction into different missing data mechanisms and patterns.

2.4.1 Missing data mechanisms

Why data are missing can have multiple reasons, and different mechanisms may lead to different consequences in statistical inference. Rubin and Little (Little & Rubin, 1987; Rubin, 1976) introduced a typology and theoretical framework of missing data mechanisms that describes the relationship between the observed and the missing data. The missing data mechanism is generally not empirically testable from the observed data alone and assumptions about the mechanism must therefore be theoretically justified (Enders, 2022, pp. 14–17).

Missing completely at random (MCAR). Considering the hypothetical complete data matrix $\mathbf{Y}^3$, $\mathbf{Y}_{com}$ is partitioned into observed, $\mathbf{Y}_{obs}$, and missing parts, $\mathbf{Y}_{mis}$, and $\mathbf{R}$ denotes the missingness⁴. Missing values are considered to be MCAR if the distribution of missingness does not depend on the data (Schafer & Graham, 2002):

$$P(\mathbf{R} \mid \mathbf{Y}_{com}) = P(\mathbf{R})$$

³Here, $\mathbf{Y}$ represents the complete data set, including both outcome variables and predictors.

⁴ $\mathbf{R}$ is treated as a set of random variables that indicate for each value if it is missing or not (Schafer & Graham, 2002).

In other words, no variable (observed or unobserved) can predict missingness and every individual in every group has the same chance of missing values (Enders, 2022, p. 6).

Missing at random (MAR). The MAR mechanism describes the case when the probability of missingness does depend on observed data, but not on missing values (Enders, 2022, p. 8):

$$P(R | Y_{com}) = P(R | Y_{obs})$$

Missingness is therefore not random, as the name would suggest, but *conditionally random*: After controlling for the observed data, the missingness is purely random (Enders, 2022, p. 8). For example, younger students may be more likely to skip the question on weekly practice duration, for instance due to reduced concentration toward the end of the questionnaire. If age is fully observed and included in the analysis model, the MAR mechanism may be plausible.

Missing not at random (MNAR). If missingness depends on unobserved values, this is called a missing not at random process⁵:

$$P(R | Y_{com}) = P(R | Y_{obs}, Y_{mis})$$

Two individuals in the same group with identical scores in all observed variables therefore no longer have the same chance of missing values, because this chance depends on the unavailable, missing information itself (Enders, 2022, p. 11). For example, students who practice less may be more likely to skip the question on weekly practice duration. If no observed variables can account for this behavior, missingness depends on the practice duration itself, which represents a MNAR mechanism.

While these mechanisms also apply to multilevel data, the situation becomes more complex because missing values can occur in the outcome, in level-1 predictors, in level-2 predictors, or even in the grouping variable (or combinations of these, Van Buuren, 2018).

2.4.2 Methods for handling missing data

Many estimation procedures require complete data. Therefore, missing data must be handled explicitly before model estimation. If researchers do not actively specify a method, statistical software often defaults to using only complete observations (Lüdtke et al., 2007). This simple ad hoc procedure is called *listwise deletion* (LD) or *complete case analysis* and is a widely used approach to handling missing data. Cases with at least one missing value are discarded, and the model is estimated using only complete observations (Graham, 2009). In multilevel settings, missingness on the group level implies that all individuals in this group are excluded as well.

Listwise deletion has two major disadvantages (Enders, 2022, p. 24):

- Because partial data is unused, statistical power is reduced.
- If the missing data mechanism is not MCAR, parameter estimates may be highly biased.

LD can still be adequate under some conditions (Enders, 2022, p. 25; Graham, 2009), but in already small samples, additional data deletion should be avoided as power is already limited (McNeish, 2017a). More modern and better-performing methods are available and will be introduced in the following sections. The approaches mentioned

⁵The special case $P(R | Y_{com}) = P(R | Y_{mis})$ is referred to as *focused MNAR* (Enders, 2022, p. 11).

here generally rely on the assumption that data are MAR⁶ and can be broadly specified into two classes:

1. *Model-based approaches* to missing data specify a statistical model for the complete data and base inference on the likelihood or posterior distribution implied by that model. Missing values are handled directly within the estimation procedure (Little & Rubin, 2020, p. 25).
2. *Imputation-based approaches* are two-fold: First, substitutions for the missing values are estimated, leading to complete data sets, which are then analyzed using standard methods.

2.4.2.1 Multiple Imputation (MI)

Multiple imputation is an imputation-based method, as the name suggests. The basic idea is to replace missing values with plausible draws (“imputations”) from their predictive distribution. The substantive analysis model is then fitted to the imputed data. In order to include the uncertainty due to missing values, in MI several imputations for each missing observation are generated. By incorporating the variance between these imputations, the additional uncertainty can be captured (Enders, 2022, p. 283). The analysis of incomplete data is separated into three steps: imputation, analysis, and pooling.

2.4.2.1.1 Creating imputations

Imputations can be understood as “predicted values plus noise” (Enders, 2022, p. 266). Multiple imputation is closely related to Bayesian data augmentation techniques, in which missing values are treated as additional unknown quantities (Tanner & Wong, 1987). From the joint distribution of regression parameters θ and the missing values $P(\theta, Y_{mis} | Y_{obs})$, parameters and imputations are simulated iteratively by an MCMC algorithm (Enders, 2022, p. 190). While during this process many thousand posterior draws are generated, only a small number (usually between $m = 20$ and $m = 100$ is recommended, Graham et al., 2007) of imputations is saved, leading to m complete data sets. The posterior draws of the parameters are discarded (Gelman et al., 2025, p. 451).

The distribution of missing values (i.e., the imputation model) is either conceptualized jointly for all variables (*joint imputation*) or individually (*fully conditional specification*). Both approaches are also suitable for multilevel data structures (Enders et al., 2016). In joint modeling, a single multivariate distribution for all variables with missing values is specified, often the multivariate normal distribution in case of continuous variables (Enders, 2022, p. 263). *Fully Conditional Specification* approximates the joint distribution by cycling through a sequence of univariate models for each variable with missings, conditional on the others. This approach is suitable if many ordinal or nominal variables are included in the data, as for every variable the researcher specifies a separate imputation model (Lüdtke et al., 2007).

Compatibility of imputation and analysis model. A crucial point in the choice of a valid imputation model is compatibility with the focal analysis model that is used later on. That is, the imputation model must reflect the key structural features of this analysis model (Grund et al., 2018b). If it fails to do so, parameter estimates may become biased because the imputed values no longer preserve the structure of the data. Consider, for example, the random intercept model with group means (Equation 2.4). In this setting, the imputation model must account for the nested data structure (i.e.,

⁶Methods for dealing with MNAR processes are not discussed in this thesis.

random effects at the group level) and allow for different effects at the individual and the group level (Grund et al., 2018b).

Auxiliary variables. Importantly, the imputation model is allowed to be “richer” than the analysis model. Including additional effects or variables that are not part of the analysis model as *auxiliary variables*⁷ is considered beneficial for parameter estimation (Collins et al., 2001; Schafer, 2003). These variables could either be a cause of missingness, thereby making the assumption of missing at random (MAR) more plausible. Furthermore, they may correlate with variables that contain missing values, which increases the precision of the imputations and reduces bias in parameter estimates (Collins et al., 2001).

Example

In the musical self-concept study, the years of teaching experience of the ensemble teachers might also be available. Although this variable is not part of the analysis model, it may correlate with practice behavior or musical self-concept and can therefore serve as an auxiliary level-2 variable.

In this thesis only the joint imputation will be considered. The *multivariate empty model* represents a suitable joint imputation model for the random intercept analysis model (Grund et al., 2016b). Revisiting the example from the previous chapter, an imputation model with one auxiliary variable can be written as

$$\begin{pmatrix} Y_{ij} \\ X_{ij} \\ W_j \end{pmatrix} = \begin{pmatrix} \beta_{0Y} \\ \beta_{0X} \\ \beta_{0W} \end{pmatrix} + \begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} + \begin{pmatrix} e_{Yij} \\ e_{Xij} \\ 0 \end{pmatrix} \quad (2.6)$$

Here, W_j represents a level-2 auxiliary variable and therefore has no level-1 residual term. The model preserves the clustered data structure by allowing random intercepts at the group level.

The random effects and residuals are assumed to follow multivariate normal distributions

$$\begin{pmatrix} u_{Yj} \\ u_{Xj} \\ u_{Wj} \end{pmatrix} \sim N(0, \Psi), \quad \begin{pmatrix} e_{Yij} \\ e_{Xij} \end{pmatrix} \sim N(0, \Sigma)$$

where the covariance matrix Ψ captures the between-group variability and covariation of the random intercepts and Σ represents the within-group covariance structure of the level-1 residuals:

$$\Psi = \begin{bmatrix} \psi_y^2 & \psi_{yx} & \psi_{yw} \\ \psi_{yx} & \psi_x^2 & \psi_{xw} \\ \psi_{yw} & \psi_{xw} & \psi_w^2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_y^2 & \sigma_{yx} \\ \sigma_{yx} & \sigma_x^2 \end{bmatrix}$$

2.4.2.1.2 Analyzing the completed data sets

After the imputation step, researchers are in the comfortable position to have m completed data sets that can be analyzed using standard frequentist methods. To each

⁷Auxiliary variables are referred to as variables whose only purpose it is to improve the missing data imputation (Collins et al., 2001).

of the data sets, the substantive analysis model of interest (e.g., the random intercept model) is now fitted. Each analysis proceeds according to the standard estimation procedure described in Section 2.2, up to the computation of parameter estimates and their standard errors. Hypothesis testing is not conducted at this stage. The result of this step is a set of m parameter estimates and corresponding standard errors.

2.4.2.1.3 Pooling the results

These m estimates and standard errors are aggregated into a single set of results following Rubin's pooling rules (Rubin, 1987).

He defined the point estimates of effects as the arithmetic average of the m estimates:

$$\hat{\theta} = \frac{1}{m} \sum_{k=1}^m \hat{\theta}_k$$

$\hat{\theta}$ is the pooled point estimate, and $\hat{\theta}_k$ is the estimate from data set k .

For the pooled standard errors, both the within-imputation variance (that is based on a complete data set) and the between-imputation variance (that adds the uncertainty due to missing values) are combined. The within-imputation variance is the arithmetic average of the imputation's variances (standard errors squared):

$$\bar{V}_w = \frac{1}{m} \sum_{k=1}^m SE_k^2$$

The between-imputation variance is defined as the variance of point estimates between imputations that is due to uncertainty caused by the missing values:

$$V_b = \frac{1}{m-1} \sum_{k=1}^m (\hat{\theta}_k - \hat{\theta})^2$$

The total pooled standard error combines the within-imputation variance $\bar{V}_w$ and the between-imputation variance V_b :

$$SE = \sqrt{\bar{V}_w + V_b + \frac{V_b}{m}}$$

The third term in this expression acts as a correction factor for using a finite number of imputations in the Monte Carlo simulation (Enders, 2022, p. 284). As the number of imputations m increases, this correction term approaches zero. As the between-imputation variance is a measure of increase in uncertainty due to missing values, the influence of missing data on the variance of estimates can be expressed proportionally, e.g. as the relative increase of variance (RIV) due to missing values:

$$RIV = \frac{V_b + \frac{V_b}{m}}{\bar{V}_w}$$

With an appropriate point estimate and standard error, the remaining step (see Figure 2.2) is the choice of the sampling distribution. Rubin proposed a calculation method for degrees of freedom of the t-distribution, that is based on the *RIV* and the

number of imputations m :

$$df_R = (m - 1) \left(1 + \frac{1}{RIV^2} \right)$$

This formula assumes that the degrees of freedom of the complete data are effectively infinite (Barnard & Rubin, 1999). In smaller samples, the classic Rubin formula therefore tends to produce degrees of freedom that are too large, sometimes even larger than the degrees of freedom of the complete data. Barnard and Rubin (1999) thus proposed a small-sample-adjusted degrees-of-freedom calculation for pooled parameters. Their formula downweights the complete-data degrees of freedom to account for the missing values. As the complete-data degrees of freedom cannot easily be obtained for multilevel analyses, a suitable degrees-of-freedom approximation must be chosen for the analysis. Possible approaches include the methods described in Section 2.3.0.3, such as the Kenward–Roger correction or the simple $m - l - 1$ rule.

2.4.2.2 Full Information Maximum Likelihood (FIML)

Full information maximum likelihood (FIML) is a model-based approach in which parameter estimates are obtained directly by maximizing the likelihood of the observed data (Enders, 2022, p. 98). Under the assumption that data are missing at random (MAR), FIML yields unbiased and efficient parameter estimates (McNeish, 2017a). Incorporating auxiliary variables is less straightforward in FIML than in multiple imputation. Nevertheless, several approaches have been proposed to include auxiliary information in FIML estimation, such as the saturated correlates model or the extra dependent variable approach (Graham, 2003), as well as the factored regression framework described below for the Bayesian missing-data approaches.

For the random intercept model with group-mean effects considered in this thesis, FIML is not always well suited when missingness occurs in predictor variables. In such cases, maximum likelihood estimation requires additional distributional assumptions about the distribution of the predictors in order to incorporate them into the likelihood function (Grund et al., 2018b). In multilevel models with manifest group means, these assumptions may lead to biased estimates of between-group effects (Grund et al., 2019).

2.4.2.3 Fully Bayesian Estimation

The Bayesian framework introduced in Section 2.3.1 can naturally be applied to missing-data problems. In contrast to the two-step procedure of multiple imputation, Bayesian missing-data analysis directly uses the posterior distribution of the model parameters conditional on the observed data for statistical inference. As described in Section 2.4.2.1.1, missing values are treated as additional unknown quantities. To simulate these values during estimation, a model for the variables with missing data must be specified. A convenient way to specify the joint distribution of the outcome and the variables with missing values is to apply the probability chain rule and express the joint distribution as a product of conditional distributions, which is referred to as *sequential specification* or *factored regression specification* (Lüdtke et al., 2020):

$$g(y, x; \gamma) = g_y(y|x; \gamma_y) \times g_x(x; \gamma_x)$$

where $\gamma = (\gamma_y, \gamma_x)$ denotes the vector of all model parameters. Here, the joint density is decomposed into two components. The first term $g_y(y | x; \gamma_y)$ corresponds to the

analysis model, describing the conditional distribution of the outcome given the covariate X . The second term $g_x(x; \gamma_x)$ specifies a model for the distribution of the covariate X . Each conditional distribution in a sequential modeling framework is typically specified as a regression model assuming normally distributed residuals. Consequently, the implied joint distribution is constructed from a sequence of conditional models, each imposing its own distributional assumptions.

An advantage of this formulation is its flexibility: it allows the inclusion of variables with different measurement scales and facilitates the use of auxiliary variables that are not part of the substantive analysis model (Daniels et al., 2014; Erler et al., 2016). For example, auxiliary variables such as W may be included to improve the imputation of missing values while the substantive model of interest is still estimated without W . For this purpose, auxiliary variables can be placed in the first position of the factored regression, so that the analysis variables predict the auxiliary variables, but not vice versa (Enders, 2022, p. 215):

$$g(y, x, w; \gamma) = g_w(w|y, x; \gamma_w) \times g_y(y|x; \gamma_y) \times g_x(x; \gamma_x) \quad (2.7)$$

Because all missing values are resampled in each iteration, the uncertainty due to missing data is automatically included in the posterior distribution (Erler et al., 2016).

2.4.3 Present study

Both challenges considered here — small sample sizes and missing data — often co-occur in empirical research, but missing-data methods have rarely been investigated in small-sample settings. For multiple imputation, some studies suggest that small-sample adjustments applied during pooling may improve inferential performance in both single-level and multilevel contexts (McNeish, 2017a, 2018). However, these adjustments have not been compared to standard inferential approaches based on unadjusted degrees of freedom, and the models considered did not incorporate contextual predictors.

Fully Bayesian estimation may offer a promising model-based alternative to MI, as it naturally accommodates both missing predictor values and small sample sizes. It is often recommended in more complex multilevel models, for example in the presence of nonlinearities or interaction effects (Depaoli & Clifton, 2015; Enders et al., 2020; Muthén & Asparouhov, 2012). However, it may also represent a suitable alternative to likelihood-based approaches such as FIML in simpler models with contextual effects, where the handling of missing predictor variables and the construction of group means may lead to biased estimates (Grund et al., 2019).

To evaluate this, the present study uses a simulation design to compare methods for handling missing data in a predictor variable in a small-sample multilevel random intercept model. In particular, it examines whether applying small-sample-adjusted degrees-of-freedom approximations after multiple imputation improves inferential performance, and whether fully Bayesian estimation provides a suitable alternative to imputation-based or likelihood-based approaches.

Simulation studies can be understood as computer experiments (Morris et al., 2019), in which data with known characteristics are generated and then analysed using the methods of interest. Because the true parameter values are known, the resulting estimates can be compared with the truth, allowing the performance of the methods to be evaluated (Siepe et al., 2024). The simulation design and evaluation criteria are described in the following chapter.

Chapter 3

Methods

Planning of the simulation study followed the ADEMP structure proposed by Siepe et al. (2024) and will be reported based on their guidelines.

3.1 Data generation mechanism

3.1.1 Data-generating model

The data-generating model was a two-level random intercept model:

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + \gamma_{02}W_j + u_{0j} + e_{ij} \quad (3.1)$$

Y_{ij} , X_{ij} and W_j were parameterized to have a mean of zero and a variance of 1 on the population level.

First, between- and within-components of X_{ij} were simulated, independent of each other and with variances chosen to achieve a prespecified intraclass coefficient (see Section 3.1.3):

$$\begin{aligned} X_{ij} &= u_{Xj} + e_{Xij}, \\ u_{Xj} &\sim N(0, \tau_x^2), \quad e_{Xij} \sim N(0, \sigma_x^2) \end{aligned}$$

with $\tau_x^2 = ICC_x$ and $\sigma_x^2 = 1 - ICC_x$, respectively.

X_{ij} was then group-mean-centered.

The group-level covariate W_j was generated to have a prespecified correlation with the group mean of X ($\rho_{\bar{X}W}$) using a linear model¹:

$$W_j = a\bar{X}_j + e_{Wj} \quad e_{Wj} \sim N(0, \sigma_w^2)$$

Finally, the random effects on the group and the individual level were generated as

$$u_{0j} \sim N(0, \psi^2), \quad e_{ij} \sim N(0, \sigma^2)$$

and independent of each other.

Individual and group residual variances of Y_{ij} were set to achieve a prespecified residual intraclass correlation and $Var(Y_{ij}) = 1$, see Appendix A.

¹Under the constraint $Var(W_j) = 1$, it follows that $\sigma_w^2 = 1 - \rho_{\bar{X}W}^2$ and $a = \frac{\rho_{\bar{X}W}}{\sqrt{\tau_x^2 + \frac{\sigma_x^2}{n}}}$, see Appendix A for details.

TABLE 3.1: Simulation conditions for the Data-Generating Model and the Generation of Missing Values

Data conditions	
N_1	10
N_2	15 / 30 / 60
ICC of X and Y	.1 / .3
Correlation $\bar{X}W$	.3
Model parameters	
Effect of $(X_{ij} - \bar{X}_{.j})$	.2
Effect of $\bar{X}_{.j}$	0 / .4
Effect of W_j	0
Missing values	
Pattern	$X \sim W$
Mechanism	MCAR / MAR
Proportion	.3

3.1.2 Missing data generation

After simulation of the complete dataset, values of the X -variable have been deleted to simulate missingness. A ‘missingness tendency’ r was simulated for each X -value according to

$$r_{ij} = \alpha + \beta W_j + e_{ij}, \quad e_{ij} \sim N(0, 1 - \beta^2)$$

r_{ij} is normally distributed. Observations with $r_{ij} > 0$ were set to missing.

The intercept α controls the proportion of missingness in X , whereas the slope β determines the strength of the missing-at-random mechanism. When $\beta = 0$, missingness is completely at random (MCAR). With increasing β , the missingness mechanism becomes more dependent on the W -values; with $\beta = 1$, missingness would be completely deterministic.

For the present simulation, $\beta = 0.3$ was used for MAR conditions, corresponding to approximately 9% explained variance.

3.1.3 Factors of the data-generating mechanism

A summary of simulation conditions is provided in Table 3.1.

The number of Level-2 units was manipulated across conditions ($N_2 = 15, 30, 60$), while the number of Level-1 observations was kept constant ($N_1 = 10$). The between-cluster effect of $\bar{X}_{.j}$ was varied ($\gamma_{01} = 0, .40$), whereas the within-cluster effect of $(X_{ij} - \bar{X}_{.j})$ was held constant at .20 and the effect of W_j was set to zero. This results in conditions with either a negative or a positive context effect (i.e., $\gamma_{10} - \gamma_{01}$) of X in the population model. The ICC of X and the residual ICC of Y were varied ($\rho_X = \rho_{Y|X} = .10, .30$). Missing data were generated on X with a varying missingness mechanism (MCAR or MAR with a strength of $\beta = .30$), with a constant missingness rate of 30%. The factors were combined fully factorially, creating $3 \times 2 \times 2 \times 2 = 24$ conditions.

Taken together, the simulation design targets conditions that are particularly challenging for multilevel estimation (small Level-2 sample sizes and low ICCs), while also covering less challenging scenarios for comparison. The between-group effect of $\bar{X}_{.j}$ is varied to evaluate method performance under a null versus a substantial effect.

3.2 Estimands of the focal analysis model

The focal analysis model is

$$Y_{ij} = \gamma_{10}(X_{ij} - \bar{X}_{\cdot j}) + \gamma_{01}\bar{X}_{\cdot j} + u_{0j} + e_{ij} \quad (3.2)$$

The primary target is the Level-2-effect, γ_{01} . Additionally, the Level-1-effect γ_{10} was evaluated.

3.3 Methods of Missing Data Handling

Four methods will be compared, alongside the benchmark analysis of Complete Data (CD).

3.3.1 Listwise Deletion (LD)

Listwise Deletion was used as a standard reference method for missing data handling. Cases with missing X -values were excluded prior to analysis. The group means $\bar{X}_{\cdot j}$ were then calculated only from the remaining observations. Then, group-mean centering was applied.

3.3.2 Multiple Imputation (MI) with standard (MI-R) and adjusted (MI-a) degrees of freedom

For both multiple imputation methods, a joint modeling approach was employed. The imputation model was the multivariate empty model specified before (Equation 2.6), with W_j as an auxiliary variable on level 2. Imputation was executed with the R package `mitml` (Grund et al., 2015) which provides an interface to the `jomo` package (Quartagno & Carpenter, 2023). The algorithm was run with 2000 burn-in-iterations and 3000 post-burn-in iterations, saving every 300th imputation, resulting in 10 imputations.

For MI-R, degrees of freedom for calculating the test statistic were calculated with the software standard (Rubin's rules, assuming infinite degrees of freedom in complete data). For MI-a, complete data degrees of freedom were calculated with $df_{\gamma_{10}} = M - r - 1$ (M : total number of Level-1 units; r : total number of predictors, in the present case 2) and $df_{\gamma_{01}} = N_2 - q - 1$ (N_2 : number of Level-2 units; q : number of Level-2 predictors, here: 1).

3.3.3 Fully Bayesian Analysis

A factored regression specification with the auxiliary variable W was used within the Bayesian framework that corresponds to the model described in Equation 2.7. The burn-in period was set to 1000 iterations, followed by 3000 post-burn-in iterations. The model was estimated using the package `mdmb` (Robitzsch & Luedtke, 2024) with its default noninformative priors.

3.3.4 Extracted quantities

For complete data, listwise deletion and both multiple imputation methods, point estimates of γ_{10} and γ_{01} as well as their standard errors and 95% - confidence intervals were extracted. For the fully Bayesian method, the posterior means of γ_{10} and γ_{01} were used as point estimates along with posterior standard deviations and 95% credible

intervals. For comparability with the frequentist approaches and in line with common simulation practice, these posterior summaries were treated analogously to frequentist estimates and intervals (Morris et al., 2019).

3.4 Performance measures

Performance was evaluated with standard performance measures for simulation studies (Morris et al., 2019; Siepe et al., 2024); namely bias, coverage, standard deviation of the estimates across simulation replications and the average width of confidence/credible intervals. Monte Carlo uncertainty of the estimated performance measures was quantified using Monte Carlo Standard Errors (MCSE), using the formulas provided in Siepe et al. (2024). For each condition, 1000 simulation replications were conducted, in line with comparable simulation studies (Enders et al., 2016; Grund et al., 2018b) and resulting in sufficiently small MCSEs for coverage. All simulations were conducted in R (version 4.5.2, R Core Team, 2025) using a reproducible workflow (e.g., fixed random seeds and controlled package environments). The full code and all relevant materials to reproduce the results are publicly available via Zenodo (<https://doi.org/10.5281/zenodo.19545308>)².

²this is only a test release right now

Chapter 4

Results

Convergence diagnostics for the MCMC algorithm were evaluated under a challenging condition (low ICC and small Level-2 sample size) and indicated satisfactory convergence with no problematic autocorrelation (see Appendix B for details). Monte Carlo uncertainty was quantified using intervals of ± 2 Monte Carlo standard errors (MCSEs), approximately corresponding to 95% confidence intervals for the estimated performance measures. For multiple imputation, results for bias and empirical standard deviations are identical for the MI-R and MI-a variants, as both approaches are based on the same point estimates and differ only in the estimation of standard errors and degrees of freedom.

For the individual-level effect γ_{10} , bias was consistently negligible relative to the corresponding Monte Carlo standard errors, and coverage, empirical standard deviations and confidence/credible interval widths were comparable across methods and closely aligned with the corresponding complete-data estimates. Results are therefore reported here only for the group-level effect γ_{01} . Full results for all parameters of interest and performance measures are provided in Appendix C.

4.1 Bias of point estimation

The findings for γ_{01} are depicted in Figure 4.1 for all conditions with missingness at random (MAR). Results under MCAR followed very similar patterns and are therefore not shown separately. A bias of 10% of the true parameter value is considered substantial for non-zero effects. For a true null effect, bias is considered substantial if it lies outside the interval of ± 2 Monte Carlo Standard Errors (MCSE).

Listwise deletion resulted in substantial bias, especially in conditions with low ICC. For a true null effect, estimates were positively biased, with a maximum of 0.042. For a positive between-group effect, estimates were negatively biased, with a maximum of -0.050. The magnitude of bias was similar across sample sizes and did not differ markedly between MAR and MCAR conditions.

Multiple imputation resulted in larger bias than listwise deletion in several low-ICC conditions with a substantial true effect, where bias ranged from -0.044 to -0.078, corresponding to approximately 10-20% of the true effect¹. In all other conditions, bias estimates for multiple imputation were small relative to their Monte Carlo standard errors (all $|\text{bias}| \leq 0.02$).

In contrast, the Bayesian approach showed only small bias across all simulation conditions except the most challenging conditions ($N_2 = 15$, $ICC = .1$, $\gamma_{01} = .4$),

¹However, in a sensitivity analysis, the use of alternative weakly informative priors in these conditions consistently reduced bias by approximately 30–60% (see Appendix D for details).

where bias was -0.039 (MAR) and -0.036 (MCAR). These values remain below 10% of the true effect. All other biases were small ($|\text{bias}| \leq 0.015$).

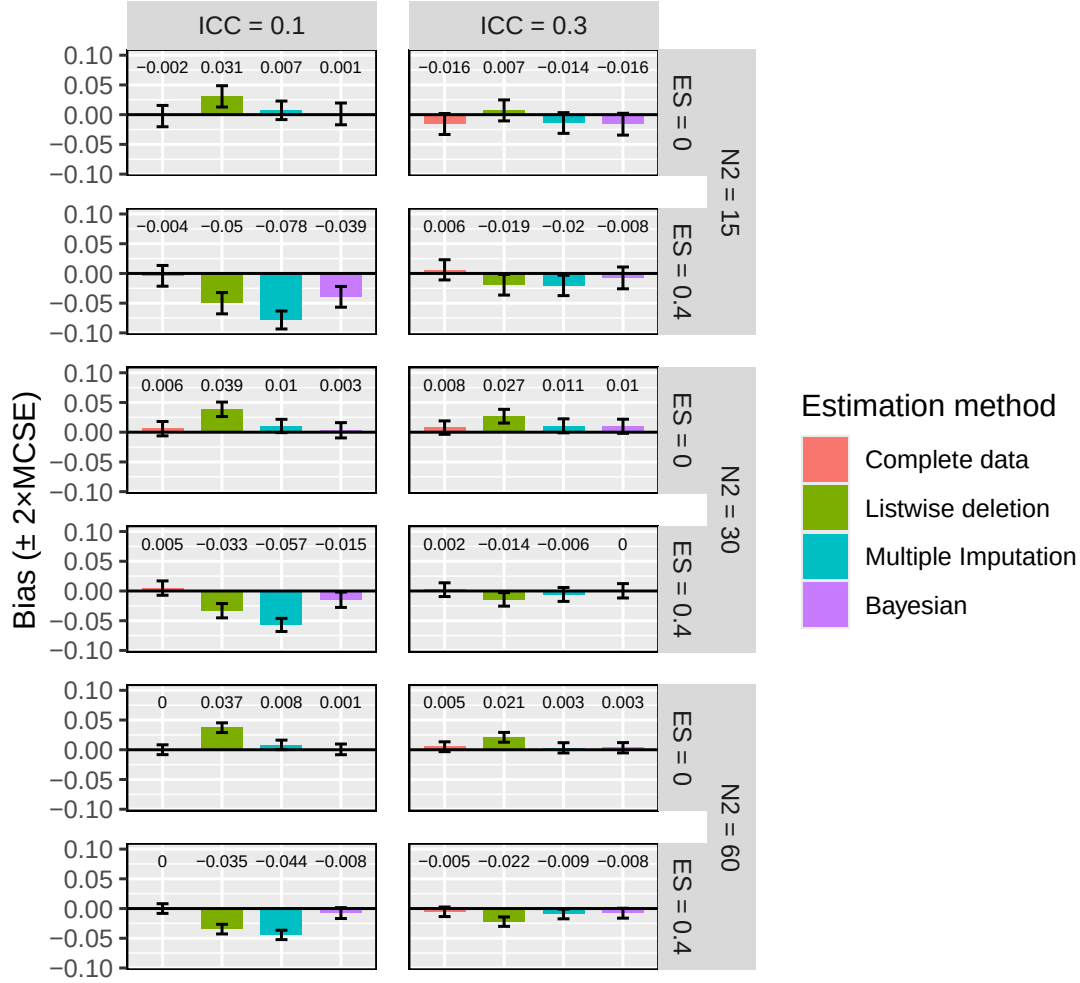


FIGURE 4.1: Bias of the between-group effect (γ_{01}) across simulation conditions under missing at random (MAR). Error bars represent ± 2 Monte Carlo standard errors. ES = Effect size.

4.2 Coverage of confidence/credible intervals

Coverage of the confidence and credible intervals for γ_{01} is shown for $N_2 = 15$ in Figure 4.2, as larger sample sizes exhibited similar but attenuated patterns. Overall, coverage rates were close to the nominal level (95%) for $N_2 = 30$ and $N_2 = 60$ across all methods.

Listwise deletion yielded coverage rates close to the nominal level in all conditions. For multiple imputation, coverage in the MI-R variant was close to the nominal level in high-ICC conditions, with a tendency toward overcoverage in low-ICC conditions. For $ICC = .1$, coverage ranged from 0.952 ($N_2 = 30$, $\gamma_{01} = .4$, MAR) to 0.971 ($N_2 = 60$, $\gamma_{01} = 0$, MCAR). The approach using small-sample adjusted degrees of freedom resulted in increased overcoverage compared to the standard degrees of freedom, particularly for $N_2 = 15$. Overcoverage was again more pronounced in low-ICC conditions, ranging from 0.960 ($N_2 = 60$, $\gamma_{01} = .4$, MCAR) to 0.986 ($N_2 = 15$, $\gamma_{01} = 0$, MCAR) for $ICC = .1$.

The Bayesian approach exhibited overcoverage for $N_2 = 15$, ranging from 0.963

($\gamma_{01} = 0$, MCAR, $ICC = .1$) to 0.985 ($\gamma_{01} = .4$, MCAR, $ICC = .1$). This method yielded the highest (most conservative) coverage rates at the smallest Level-2 sample size, except in the condition with MCAR, $ICC = .1$, and $\gamma_{01} = 0$. With increasing Level-2 sample size, this pattern attenuated. For $N_2 = 30$, coverage estimates were largely within the bounds of Monte Carlo error, and for $N_2 = 60$, they fell entirely within these bounds across all conditions.

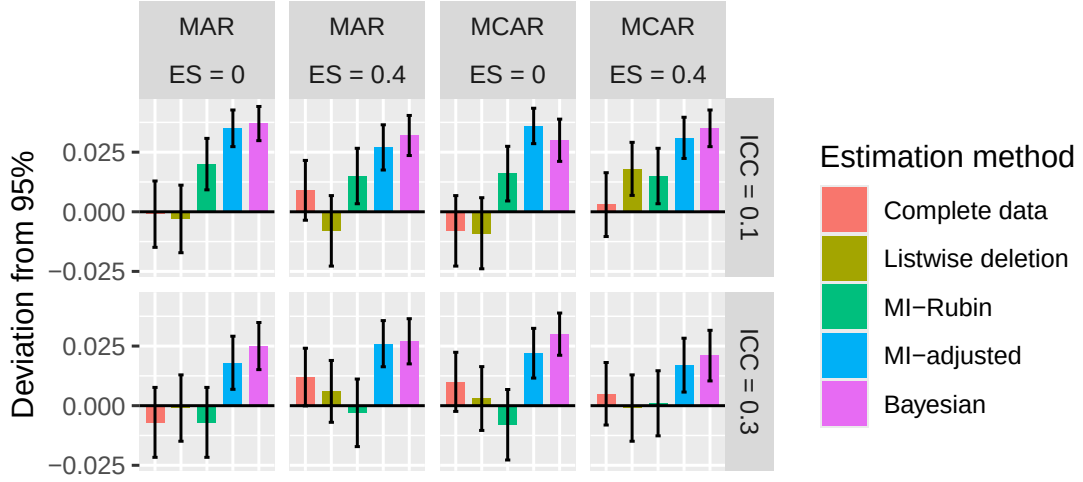


FIGURE 4.2: Deviation from nominal (95%) coverage of confidence/credible intervals across methods and simulation conditions for $N_2 = 15$. Error bars represent ± 2 Monte Carlo standard errors. MI = Multiple Imputation; ES = Effect size.

4.3 Standard deviation of point estimates

Standard deviations of point estimates across conditions and methods are illustrated in Figure 4.3 for missingness at random (MAR), as differences between missingness mechanisms were small.

Listwise deletion produced standard deviations very close to those obtained under complete-data analysis, ranging from 0.124 ($N_2 = 60$, MCAR, $\gamma_{01} = 0.4$) to 0.300 ($N_2 = 15$, MCAR, $\gamma_{01} = 0.0$). For multiple imputation, standard deviations were comparable to complete-data analysis across conditions, with a tendency toward smaller values in some low-ICC conditions. For example, for $N_2 = 15$, $\gamma_{01} = 0.0$, and MAR missingness, multiple imputation yielded a standard deviation of 0.247, compared to 0.283 under complete-data analysis.

The Bayesian approach yielded standard deviations that were generally slightly larger than those from complete-data analysis, typically by approximately 0.006 to 0.020. Standard deviations were generally smaller under high ICC conditions than under low ICC conditions in the Bayesian framework, except in the conditions with $\gamma_{01} = 0.4$ and $N_2 = 15$. In contrast, multiple imputation consistently yielded slightly larger standard deviations under high-ICC conditions. Although these differences were small, they were larger than would be expected from Monte Carlo error alone.

4.4 Width of confidence / credible intervals

Results are illustrated in Figure 4.4, as similar patterns were observed across conditions. Overall, the Bayesian approach yielded the widest credible intervals, followed by multiple imputation with adjusted degrees of freedom (MI-a). For $N_2 = 15$, the MI-R

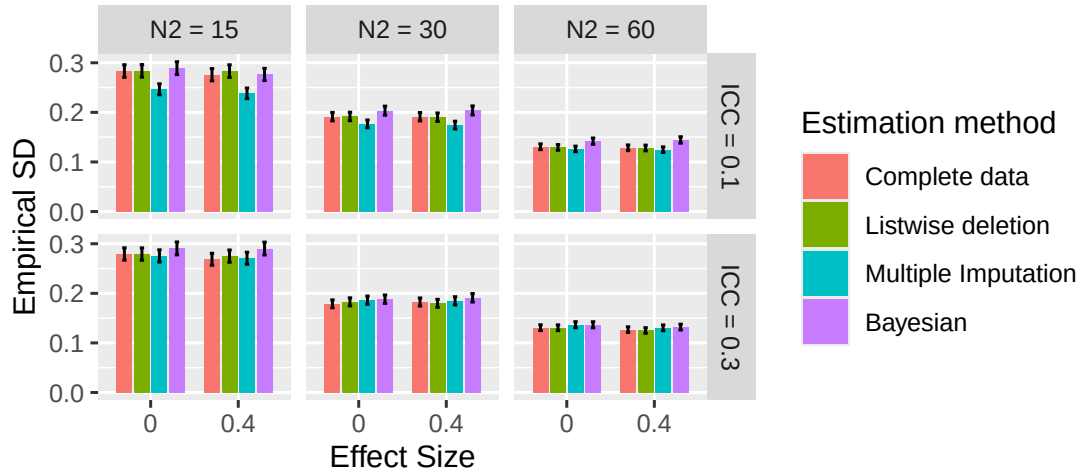


FIGURE 4.3: Empirical standard deviation of point estimates for γ_{01} for conditions with missingness at random (MAR). Error bars represent ± 2 Monte Carlo standard errors.

variant produced interval widths that were slightly smaller than those obtained under complete-data analysis (e.g., 1.115 vs. 1.170 for MCAR, $ICC = .1$, $\gamma_{01} = 0.0$). For $N_2 = 30$ and $N_2 = 60$, MI-R interval widths were slightly larger than those from complete-data analysis, while still remaining smaller than those from MI-a and the Bayesian approach. Confidence intervals from listwise deletion were similar to those obtained with complete data analysis across conditions.

At the smallest Level-2 sample size, $N_2 = 15$, differences between methods were most pronounced. For example, when $\gamma_{01} = 0.0$ and $ICC = .1$ under MCAR, interval width for γ_{01} was 1.115 under MI-R, 1.284 with MI-a and 1.456 under the Bayesian approach. With increasing Level-2 sample size, differences between methods became less pronounced, although interval widths for the Bayesian approach remained slightly larger. For instance, when $\gamma_{01} = 0.0$, $ICC = .1$, and MCAR at $N_2 = 60$, interval width was 0.517 for complete-data analysis, 0.520 for listwise deletion, 0.545 for MI-R, 0.559 for MI-a, and 0.582 for the Bayesian approach. Within the fully Bayesian approach, interval widths were generally slightly smaller under high-ICC conditions than under low-ICC conditions. In contrast, the other methods did not show a consistent ICC-related pattern.

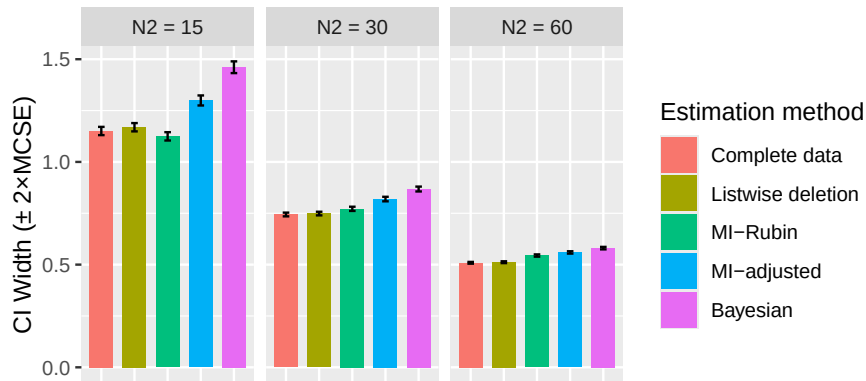


FIGURE 4.4: Width of confidence/credible intervals for $\gamma_{01} = .4$ under missingness at random (MAR) and $ICC = .1$. Error bars represent ± 2 Monte Carlo standard errors.

4.5 Summary of results

The methods showed different performance profiles for the group-level effect. Listwise deletion resulted in increased bias, while other performance measures were generally similar to complete-data results. Multiple imputation produced substantial downward bias in the estimation of the level-2 effect under conditions of low ICC and MAR missingness. At the same time, the standard variant (MI-R) yielded coverage rates close to the nominal level across conditions. Applying a small-sample correction to the degrees of freedom led to increased coverage beyond the nominal level and wider confidence intervals. The fully Bayesian approach showed small to negligible bias across conditions, but resulted in the widest intervals among all methods and a tendency toward overcoverage in the smallest sample size.

Chapter 5

Discussion

This thesis investigated the performance of two modern methods for handling missing predictor data in small-sample multilevel models, multiple imputation and a fully Bayesian approach, alongside listwise deletion as a commonly used default. It focused on a reduced setting with a single continuous predictor which was included at level 1 as an individual-level predictor and at level 2 as a manifest group-mean predictor, within a random intercept model. In addition, one auxiliary variable at level 2 was included to explain missingness under a MAR mechanism, without being directly related to the outcome. The analytical setup was designed to reflect common applied practice, in which default prior specifications provided by standard software are frequently used. There is still limited literature addressing the combined challenges of multilevel data, small sample sizes, and missing data. Against this background, focusing on a simplified setting allowed for a controlled investigation of method performance.

The results indicate a trade-off between bias and interval width when comparing multiple imputation and the fully Bayesian approach. In settings with low ICC and default prior specifications, the fully Bayesian approach may offer advantages over multiple imputation with respect to bias, with differences persisting even at comparatively larger cluster sizes in the present design. Although multiple imputation can be understood as an approximate Bayesian procedure, as it relies on draws from the posterior predictive distribution (e.g., Enders, 2022, p. 299; Schafer & Graham, 2002), the present results suggest that this connection does not necessarily translate into comparable performance in small samples. In particular, prior specification played an important role, as indicated by the sensitivity analysis. When reasonable prior information is available for level-2 variance components, incorporating this information through weakly informative priors can reduce bias in multiple imputation while holding coverage at a nominal level.

Another aspect of multiple imputation that was specifically investigated in this thesis is the behavior of inference based on uncorrected versus small-sample-adjusted degrees of freedom. The findings suggest that, in this setting, standard Rubin's rules already yield approximately valid inference, and that additional small-sample corrections may not be necessary. This is somewhat surprising, as it is commonly recommended to use an appropriate sampling distribution in small samples to avoid overly narrow confidence intervals; a recommendation which also applies to multilevel analyses (Kenward & Roger, 1997; Schaalje et al., 2002) and within multiple imputation (Barnard & Rubin, 1999). However, confidence intervals are influenced by multiple sources of uncertainty beyond the degrees-of-freedom approximation alone. In particular, REML estimation was used for analyzing the imputed data sets. Compared to ML, REML typically yields larger estimates of level-2 variance components, which have been shown to improve variance estimation in small samples (Elff et al., 2021; McNeish & Stapleton,

2016b). As these variance components directly affect the standard errors of the fixed effects, they also influence the within-imputation variance and, consequently, the total variance used for inference. This may have contributed to wider intervals and thus higher coverage. In addition, multiple imputation was performed using joint modeling via `jomo`. Previous work has shown that inverse Wishart priors with few degrees of freedom, as implemented in `jomo`, can introduce additional variability into fixed-effect estimates (Audigier et al., 2018). Overall, this finding may be reassuring for applied researchers relying on default degrees-of-freedom specifications.

In terms of implementation, both fully Bayesian estimation and multiple imputation allow the inclusion of auxiliary variables, which were also part of the present setup. In multiple imputation, their inclusion is typically well-documented and supported by standard workflows. In fully Bayesian approaches, however, incorporating auxiliary variables without altering the analysis model requires additional modeling decisions that are less explicitly supported by standardized workflows and software guidance and may therefore not be as straightforward to practitioners. Against this background, the choice between approaches is likely to depend not only on statistical performance but also on the research context and analytical goals. Multiple imputation may be advantageous in settings involving multiple analyses (e.g., multiple outcomes), where standardized workflows and comparability across analyses are desired, as completed data sets can be reused and shared. In contrast, fully Bayesian approaches may be particularly useful in cumulative research settings, where prior information can be updated across studies and evidence can be built up and integrated over time. However, these distinctions are not absolute, and both approaches can, in principle, be adapted to a wide range of analytical scenarios.

This simulation study comes with several limitations. Firstly, more complex model structures may behave differently under the investigated methods. The present study focused on a relatively simple model without nonlinear terms, interaction effects, or random slopes. Sample size requirements in multilevel models generally depend on model complexity, as more complex specifications involve a larger number of parameters that must be estimated reliably (Hox & McNeish, 2020). Therefore, what worked well in the present study might be more problematic in more complex models. In particular, when such features are present, the treatment of missing data becomes substantially more challenging (Enders et al., 2016; Grund et al., 2016a, 2018b). For multiple imputation, default imputation models often lead to incompatibility between the imputation and analysis model and, consequently, biased estimates (Bartlett et al., 2015; Kim et al., 2018; Seaman et al., 2012). Therefore, approaches that allow for substantive-model-compatible MI, such as sequential modeling, have been proposed to address these challenges (Bartlett et al., 2015; Enders et al., 2020; Grund et al., 2021; Lüdtke et al., 2020; Zhang & Wang, 2017), but these approaches have not yet been thoroughly investigated in data sets with few clusters.

Second, the present results may not generalize to more extreme small-sample settings. With very few clusters, for example fewer than 10, variance components are difficult to estimate reliably, which can affect both multiple imputation and Bayesian approaches (Hox et al., 2012; Maas & Hox, 2005; McNeish & Stapleton, 2016a). Furthermore, cluster size plays an important role for estimation and was fixed at 10 in the present study. With very few observations per cluster and unbalanced data (i.e., unequal number of observations per cluster), different patterns of results might be obtained; e.g., very small cluster sizes are connected to an upward bias in level-2 variance components if number of groups was also small (Clarke, 2008; McNeish, 2014). Empirical evidence on the effects of unbalanced cluster sizes is sparse, but existing literature suggests that

they can bias statistical inference by leading to underestimated standard errors of fixed effects, particularly when combined with heteroscedasticity and a small number of clusters (Korendijk et al., 2008; Raudenbush & Bryk, 2010, pp. 13, 281). Unbalanced data also have implications for the imputation design in contextual models. In such settings, imputation models that rely on manifest group means and don't account for cluster size may introduce additional bias (Grund et al., 2018a).

Third, prior specification was only examined in a limited sensitivity analysis for multiple imputation in the present study. In complete-data settings, previous research suggests that weakly informative priors often outperform noninformative priors in Bayesian multilevel models with few clusters (Gelman, 2006; McNeish, 2016a; Smid et al., 2020), whereas findings on data-dependent priors remain inconclusive (Depaoli, 2013; McNeish, 2016b; Van Erp et al., 2018). However, studies on the role of prior specification in the presence of missing data and few clusters remain sparse. For multiple imputation in particular, standard software priors are often used, although some findings suggest that prior choice can substantially influence results in multiple imputation as well (Audigier et al., 2018; Grund et al., 2018b). Extending this line of research to settings with missing data would therefore be an important direction for future research.

Fourth, several key modeling choices were not systematically varied in the present study. Future research could manipulate estimation methods (ML versus REML), imputation frameworks (joint modeling versus fully conditional specification), small-sample corrections (e.g., Kenward-Roger, $m - l - 1$), prior specifications and model complexity to disentangle their relative contributions to estimation performance. Such work would help clarify the behavior of multiple imputation in comparison with fully Bayesian estimation and identify the conditions under which small-sample corrections for degrees of freedom are useful, as well as those under which they may not be necessary. Existing evidence suggests that degree-of-freedom corrections might be beneficial in more complex models, such as growth models with random slopes and longitudinal missingness patterns (McNeish, 2018).

Fifth, two distinct software packages with different default prior specifications were used for multiple imputation and Bayesian estimation. Using a common software framework for both approaches, as for example possible with `mdmb` (Grund et al., 2021; Robitzsch & Luedtke, 2024), could improve comparability and help isolate differences attributable to modeling choices rather than implementation details.

Finally, a further limitation concerns the missing data mechanism. The present study assumes missingness (completely) at random, whereas the combination of few clusters and missing data should also be investigated under missingness not at random, which requires different modeling approaches (Du et al., 2022; Enders, 2025). Moreover, the present missingness pattern with missings only in one predictor variable is very simplified, as in real-world applications, missing values typically occur across multiple variables, including outcomes, level-1 and level-2 predictors, and the grouping variable (Van Buuren, 2018). Even if the joint modeling approach used in this thesis can accommodate multivariate missings, bias and efficiency of the estimates might be differently affected by different model specifications.

Overall, the present findings provide insight into the behavior of missing data methods in small-sample multilevel settings with missings in a predictor variable, but further research is required to assess their robustness under more complex and realistic conditions.

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Appendix A

Derivations of equations of data-generating model

A.1 Calculation of σ_w^2

The data-generating model for W_j is:

$$W_j = a \cdot \bar{X}_j + e_{Wj}, \quad e_{Wj} \sim N(0, \sigma_w^2) \quad (\text{A.1})$$

The variance of W_j is therefore:

$$\text{Var}(W_j) = \text{Var}(a \cdot \bar{X}_j + e_{Wj}) = a^2 \cdot \text{Var}(\bar{X}_j) + \text{Var}(e_{Wj}) \quad (\text{A.2})$$

Substituting the formula for slope calculation $a = \frac{\text{Cov}(W_j, \bar{X}_j)}{\text{Var}(\bar{X}_j)}$ into Equation [A.2](#) yields:

$$\begin{aligned} \text{Var}(W_j) &= \frac{\text{Cov}(W_j, \bar{X}_j)^2}{\text{Var}(\bar{X}_j)^2} \cdot \text{Var}(\bar{X}_j) + \text{Var}(e_{Wj}) \\ &= \frac{\text{Cov}(W_j, \bar{X}_j)^2}{\text{Var}(\bar{X}_j)} + \text{Var}(e_{Wj}) \end{aligned}$$

This term can be simplified using $\text{Cov}(A, B) = \text{Corr}(A, B) \cdot \text{SD}(A) \cdot \text{SD}(B)$:

$$\begin{aligned} \text{Var}(W_j) &= \frac{\text{Cov}(W_j, \bar{X}_j)^2}{\text{Var}(\bar{X}_j)} + \text{Var}(e_{Wj}) \\ &= \frac{(\text{Corr}(W_j, \bar{X}_j) \cdot \text{SD}(W_j) \cdot \text{SD}(\bar{X}_j))^2}{\text{Var}(\bar{X}_j)} + \text{Var}(e_{Wj}) \\ &= \frac{\text{Corr}(W_j, \bar{X}_j)^2 \cdot \text{Var}(W_j) \cdot \text{Var}(\bar{X}_j)}{\text{Var}(\bar{X}_j)} + \text{Var}(e_{Wj}) \\ &= \text{Corr}(W_j, \bar{X}_j)^2 \cdot \text{Var}(W_j) + \text{Var}(e_{Wj}) \end{aligned}$$

Thus,

$$\begin{aligned}
Var(W_j) &= Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) + Var(e_{W_j}) \\
Var(e_{W_j}) &= Var(W_j) - Corr(W_j, \bar{X}_j)^2 \cdot Var(W_j) \\
&= Var(W_j) (1 - Corr(W_j, \bar{X}_j)^2)
\end{aligned}$$

Since $Var(W_j) = 1$:

$$Var(e_{W_j}) = 1 - Corr(W_j, \bar{X}_j)^2$$

In short notation:

$$\sigma_w^2 = 1 - \rho_{\bar{X}W}^2$$

A.2 Calculation of the slope to generate W_j

$$a = \frac{Cov(W_j, \bar{X}_j)}{Var(\bar{X}_j)}$$

The variance of $\bar{X}_j$ can be derived as follows:

$$\begin{aligned}
Var(\bar{X}_j) &= Var\left(\frac{1}{n} \sum_{i=1}^n X_{ij}\right) \\
&= Var\left(\frac{1}{n} \sum_{i=1}^n (u_{Xj} + e_{Xij})\right) \\
&= Var\left(\frac{1}{n} \sum_{i=1}^n u_{Xj} + \frac{1}{n} \sum_{i=1}^n e_{Xij}\right) \\
&= Var\left(u_{Xj} + \frac{1}{n} \sum_{i=1}^n e_{Xij}\right) \\
&= Var(u_{Xj}) + Var\left(\frac{1}{n} \sum_{i=1}^n e_{Xij}\right) \\
&= Var(u_{Xj}) + \frac{1}{n^2} Var\left(\sum_{i=1}^n e_{Xij}\right) \\
&= Var(u_{Xj}) + \frac{1}{n^2} \sum_{i=1}^n Var(e_{Xij}) \\
&= \tau_x^2 + \frac{n \cdot \sigma_X^2}{n^2} \\
&= \tau_x^2 + \frac{\sigma_X^2}{n}
\end{aligned} \tag{A.3}$$

where n denotes the group size (cluster size), assumed to be constant across groups. The covariance $Cov(W_j, \bar{X}_j)$ is therefore:

$$\begin{aligned}
Cov(W_j, \bar{X}_j) &= \rho_{\bar{X}W} \cdot SD(\bar{X}_j) \cdot SD(W_j) \\
&= \rho_{\bar{X}W} \cdot \sqrt{Var(\bar{X}_j)} \cdot \sqrt{Var(W_j)} \\
&= \rho_{\bar{X}W} \cdot \sqrt{\tau_x^2 + \frac{\sigma_X^2}{n}} \cdot 1
\end{aligned}$$

A.3 Variances of random components of Y

Individual and group residual variances of Y ($Var(u_{0j}) = \psi^2$ and $Var(e_{ij}) = \sigma^2$) are derived to match a prespecified residual ICC of Y :

$$ICC_{Y|X,W} = \frac{\psi^2}{\psi^2 + \sigma^2}$$

Because $Var(Y) = 1$ and the random components are uncorrelated with the predictors,

$$\psi^2 + \sigma^2 = Var(\text{Residual}) = 1 - Var(\text{fixed})$$

where $Var(\text{fixed})$ denotes the variance explained by the fixed effects. The variance components can therefore be expressed as:

$$\begin{aligned}
Var(u_{0j}) &= ICC_{Y|X,W} \cdot (1 - Var(\text{fixed})) \\
Var(e_{ij}) &= (1 - ICC_{Y|X,W}) \cdot (1 - Var(\text{fixed}))
\end{aligned}$$

$Var(\text{fixed})$ can be calculated as:

$$Var(\text{fixed}) = Var(\gamma_{10}(X_{ij} - \bar{X}_{.j}) + \gamma_{01}\bar{X}_{.j} + \gamma_{02}W_j)$$

The terms $(X_{ij} - \bar{X}_{.j})$ and $\bar{X}_{.j}$ are uncorrelated. W_j is correlated with $\bar{X}_{.j}$ but not with X_{ij} (see Equation Equation A.1), and $Var(W_j)$ is set to 1, which yields:

$$Var(\text{fixed}) = \gamma_{10}^2 \cdot Var(X_{ij} - \bar{X}_{.j}) + \gamma_{01}^2 \cdot Var(\bar{X}_{.j}) + \gamma_{02}^2 + 2\gamma_{01}\gamma_{02}Cov(\bar{X}_{.j}, W_j)$$

Because γ_{02} is set to zero in the present design, this term can be simplified to:

$$Var(\text{fixed}) = \gamma_{10}^2 \cdot Var(X_{ij} - \bar{X}_{.j}) + \gamma_{01}^2 \cdot Var(\bar{X}_{.j}) \quad (\text{A.4})$$

The variance of $(X_{ij} - \bar{X}_{.j})$ can be derived as follows:

$$Var(X_{ij} - \bar{X}_{.j}) = Var(X_{ij}) + Var(\bar{X}_{.j}) - 2 \cdot Cov(X_{ij}, \bar{X}_{.j})$$

$Cov(X_{ij}, \bar{X}_{.j})$ can be expressed as

$$Cov(X_{ij}, \bar{X}_{.j}) = Cov\left(X_{ij}, \frac{1}{n} \sum_{i'=1}^n X_{i'j}\right)$$

where i' may be the same or a different index than i . Using the decomposition of X_{ij} introduced in Section Section 3.1.1, $X_{ij} = u_{Xij} + e_{Xij}$, the covariance term can be rewritten and expanded as

$$\begin{aligned}
\text{Cov}(X_{ij}, \bar{X}_{.j}) &= \text{Cov} \left(u_{Xj} + e_{Xij}, \frac{1}{n} \sum_{i'=1}^n (u_{Xj} + e_{Xi'j}) \right) \\
&= \text{Cov} \left(u_{Xj} + e_{Xij}, u_{Xj} + \frac{1}{n} \sum_{i'=1}^n e_{Xi'j} \right) \\
&= \text{Cov}(u_{Xj}, u_{Xj}) + \text{Cov} \left(u_{Xj}, \frac{1}{n} \sum_{i'=1}^n e_{Xi'j} \right) \\
&\quad + \text{Cov}(e_{Xij}, u_{Xj}) + \text{Cov} \left(e_{Xij}, \frac{1}{n} \sum_{i'=1}^n e_{Xi'j} \right)
\end{aligned}$$

Because of uncorrelatedness between u_{Xj} and e_{Xij} , the second and third term are zero; furthermore, $\text{Cov} \left(e_{Xij}, \frac{1}{n} \sum_{i'=1}^n e_{Xi'j} \right) = \frac{1}{n} \sum_{i'=1}^n \text{Cov}(e_{Xij}, e_{Xi'j})$. As $\text{Cov}(e_{Xij}, e_{Xi'j}) = \text{Var}(e_{Xij})$ if $i = i'$ and zero otherwise, $\frac{1}{n} \sum_{i'=1}^n \text{Cov}(e_{Xij}, e_{Xi'j}) = \frac{1}{n} \text{Var}(e_{Xij})$.

Overall, the covariance now simplifies to

$$\begin{aligned}
\text{Cov}(X_{ij}, \bar{X}_{.j}) &= \text{Var}(u_{Xj}) + \frac{1}{n} \text{Var}(e_{Xij}) \\
&= \tau_x^2 + \frac{\sigma_X^2}{n} \\
&= \text{Var}(\bar{X}_{.j})
\end{aligned}$$

Returning to $\text{Var}(X_{ij} - \bar{X}_{.j})$, this simplifies to:

$$\begin{aligned}
\text{Var}(X_{ij} - \bar{X}_{.j}) &= \text{Var}(X_{ij}) + \text{Var}(\bar{X}_{.j}) - 2 \cdot \text{Cov}(X_{ij}, \bar{X}_{.j}) \\
&= \text{Var}(X_{ij}) + \text{Var}(\bar{X}_{.j}) - 2 \cdot \text{Var}(\bar{X}_{.j}) \\
&= \text{Var}(X_{ij}) - \text{Var}(\bar{X}_{.j}) \\
&= (\tau_x^2 + \sigma_X^2) - (\tau_x^2 + \frac{\sigma_X^2}{n}) \\
&= \sigma_X^2 - \frac{\sigma_X^2}{n}
\end{aligned}$$

Inserting the derived terms into Equation A.4, $\text{Var}(\text{fixed})$ can be therefore calculated as

$$\text{Var}(\text{fixed}) = \gamma_{10}^2 \left(\sigma_X^2 - \frac{\sigma_X^2}{n} \right) + \gamma_{01}^2 \left(\tau_x^2 + \frac{\sigma_X^2}{n} \right).$$

The variances of the random components can thus be calculated like this:

$$\begin{aligned}
\psi^2 &= ICC_{Y|X,W} \cdot (1 - \gamma_{10}^2 (\sigma_X^2 - \frac{\sigma_X^2}{n}) + \gamma_{01}^2 (\tau_x^2 + \frac{\sigma_X^2}{n})) \\
\sigma^2 &= (1 - ICC_{Y|X,W}) \cdot (1 - \gamma_{10}^2 (\sigma_X^2 - \frac{\sigma_X^2}{n}) + \gamma_{01}^2 (\tau_x^2 + \frac{\sigma_X^2}{n}))
\end{aligned}$$

Appendix B

Convergence analysis for MI and fully Bayesian MCMC

Convergence behaviour was tested in one run of the most challenging condition ($N_2 = 15, \gamma_{01} = .4, \beta = .3, ICC = .1$).

B.1 Diagnostics of multiple imputation

Convergence of the imputation model was assessed using potential scale reduction factors (Gelman & Rubin, 1992). Scale reduction factors that are substantially larger than 1 indicate that convergence was not yet reached fully; an acceptable threshold can be set at 1.1 (Gelman et al., 2025, pp. 285–287), while a stricter recommendation was recently set to be 1.01 (Vehtari et al., 2021). Given the simulation setting with aggregated results over many replications, the first threshold was considered sufficient.

All values were close to 1, indicating satisfactory convergence across all parameters. A maximum $\hat{R}$ of 1.026 was observed for the intercept of X in the imputation model. Diagnostic plots are shown in Figure B.1 - B.3 for this parameter and the most critical parameters, the level-2 variances of Y and X .

Trace plots showed no signs of substantial change after burn-in. Inspection of the autocorrelation plots indicated that autocorrelation diminished quickly across increasing lags and did not persist over longer distances between samples. The distance of 300 lags between imputation that was chosen in this simulation was therefore sufficient.

B.2 Diagnostics of fully Bayesian analysis

$\hat{R}$ values were close to 1, except for the level-2 variance of X ($\hat{R} = 1.05$) and the regression coefficient of the group mean of X on W ($\hat{R} = 1.03$). Diagnostic plots of these parameters and the additionally critical parameter τ_{y0}^2 are shown in Figure B.4. Trace plots showed no drifting. Autocorrelation decreased rapidly for most parameters, although some exhibited persistent but low autocorrelation across lags. This pattern suggests slightly reduced mixing efficiency rather than substantial convergence problems (Johnson et al., 2022, pp. 149–152).

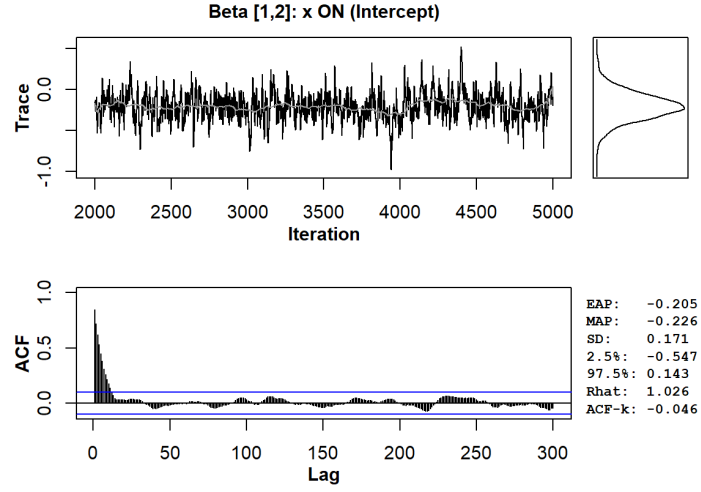


FIGURE B.1: Trace and autocorrelation plot in an exemplary simulation run for the intercept in the imputation model for X (multiple imputation).

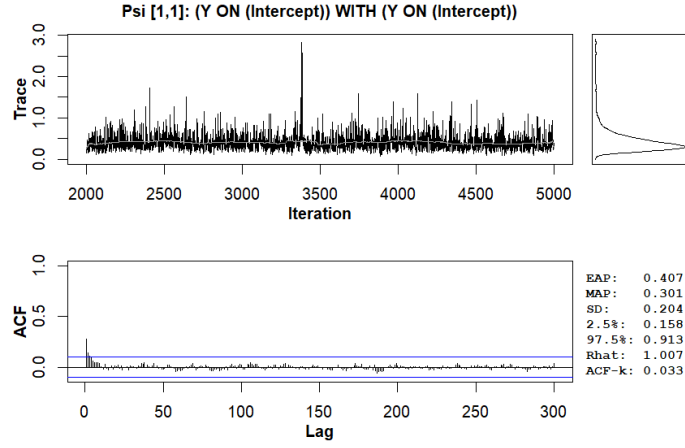


FIGURE B.2: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of Y (multiple imputation).

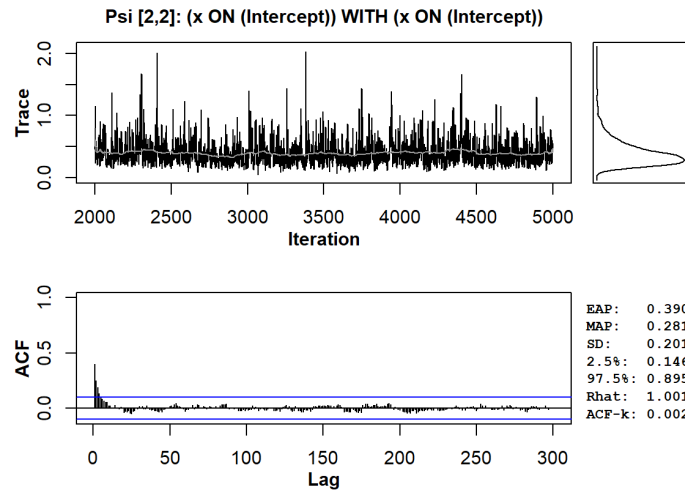


FIGURE B.3: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of X (multiple imputation).

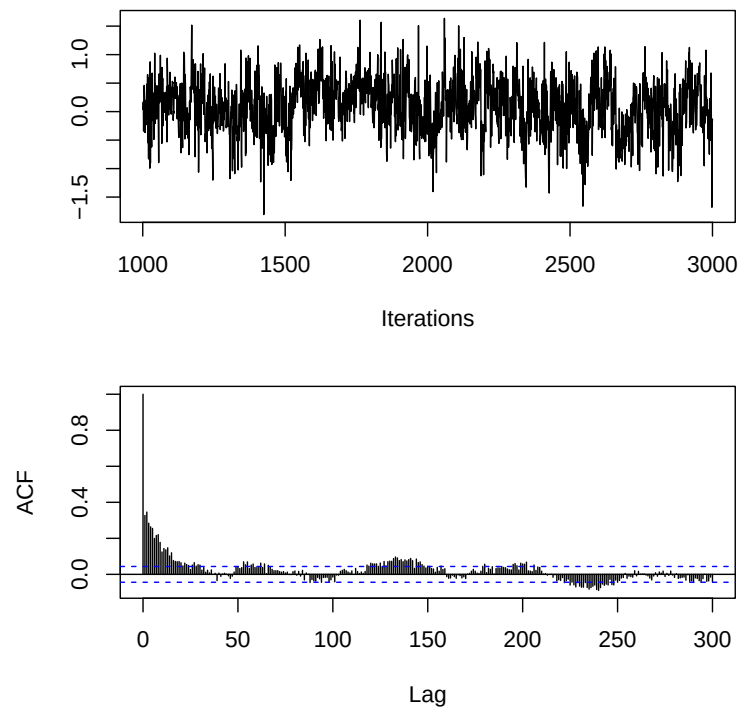


FIGURE B.4: Trace and autocorrelation plot for the effect of the group mean of X on W in the imputation model (fully Bayesian estimation).

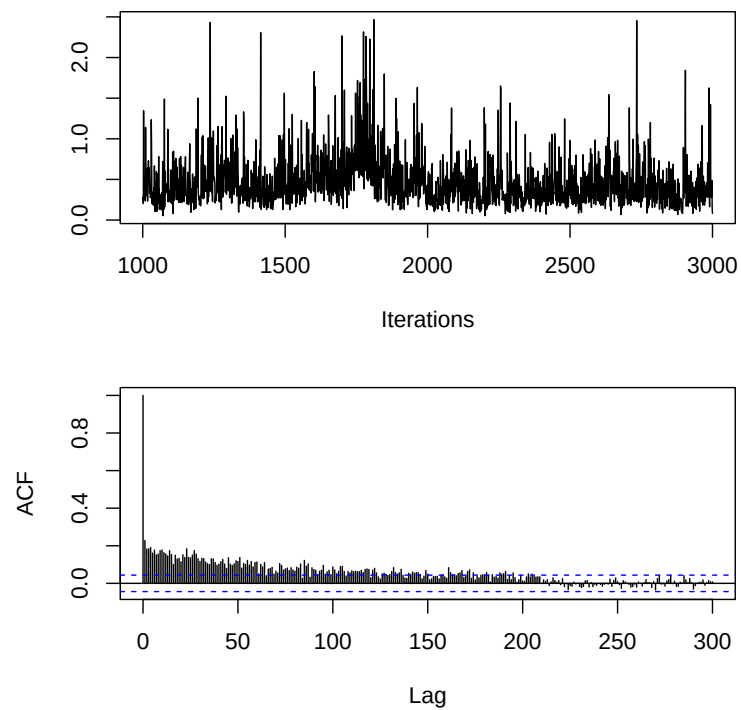


FIGURE B.5: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of X (fully Bayesian analysis)

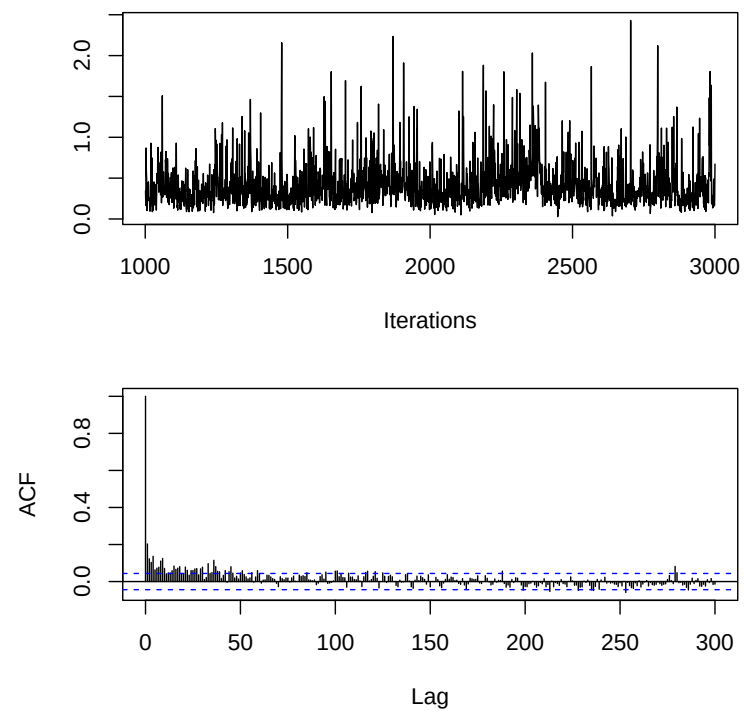


FIGURE B.6: Trace and autocorrelation plot in an exemplary simulation run for the level-2 variance of Y (fully Bayesian analysis)

Appendix C

Complete simulation results

TABLE C.1: Simulation results ($N_2 = 15$, $\gamma_{01} = 0.0$)

Parameter	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.002	0.942	0.284	1.170	0.037	0.941	0.300	1.180	0.005	0.966	0.252	1.115	0.005	0.986	0.252	1.284	0.003	0.980	0.291	1.456
γ_{10}	-0.006	0.951	0.084	0.335	-0.006	0.944	0.104	0.414	-0.008	0.943	0.103	0.409	-0.008	0.944	0.103	0.414	-0.010	0.946	0.100	0.401
ICC = 0.3																				
γ_{01}	0.000	0.960	0.272	1.174	0.012	0.953	0.280	1.177	-0.001	0.942	0.278	1.126	-0.001	0.972	0.278	1.278	-0.003	0.980	0.288	1.384
γ_{10}	0.001	0.942	0.089	0.336	0.000	0.931	0.111	0.415	-0.005	0.938	0.108	0.411	-0.005	0.939	0.108	0.416	-0.006	0.934	0.108	0.405
MAR																				
ICC = 0.1																				
γ_{01}	-0.002	0.949	0.283	1.192	0.031	0.947	0.284	1.189	0.007	0.970	0.247	1.131	0.007	0.985	0.247	1.305	0.001	0.987	0.289	1.481
γ_{10}	0.000	0.942	0.087	0.336	-0.002	0.954	0.106	0.416	-0.007	0.949	0.104	0.412	-0.007	0.950	0.104	0.418	-0.007	0.954	0.102	0.403
ICC = 0.3																				
γ_{01}	-0.016	0.943	0.279	1.175	0.007	0.949	0.279	1.175	-0.014	0.943	0.275	1.130	-0.014	0.968	0.275	1.285	-0.016	0.975	0.291	1.390
γ_{10}	0.001	0.941	0.087	0.337	-0.001	0.948	0.105	0.414	-0.004	0.943	0.102	0.411	-0.004	0.945	0.102	0.417	-0.005	0.949	0.101	0.405

Note:

Bias = average deviation from the true parameter; Cov = empirical coverage probability; SD = empirical standard deviation; CIW = mean confidence/credible interval width. MCAR = missing completely at random; MAR = missing at random. CD = complete data; LD = listwise deletion; MI-R = multiple imputation with Rubin's rules; MI-a = multiple imputation with adjusted degrees of freedom; Bayes = fully Bayesian estimation.

TABLE C.2: Simulation results ($N_2 = 15$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	-0.008	0.953	0.275	1.174	-0.038	0.968	0.269	1.183	-0.075	0.965	0.239	1.151	-0.075	0.981	0.239	1.329	-0.036	0.985	0.281	1.516
γ_{10}	0.000	0.952	0.083	0.330	0.000	0.952	0.102	0.404	-0.001	0.950	0.101	0.401	-0.001	0.953	0.101	0.407	-0.003	0.954	0.099	0.393
ICC = 0.3																				
γ_{01}	0.003	0.955	0.272	1.154	-0.013	0.949	0.272	1.156	-0.016	0.951	0.275	1.122	-0.016	0.967	0.275	1.277	-0.008	0.971	0.290	1.385
γ_{10}	0.002	0.949	0.083	0.326	0.003	0.935	0.106	0.402	-0.001	0.944	0.103	0.397	-0.001	0.948	0.103	0.402	-0.003	0.942	0.101	0.391
MAR																				
ICC = 0.1																				
γ_{01}	-0.004	0.959	0.276	1.151	-0.050	0.942	0.283	1.169	-0.078	0.965	0.238	1.124	-0.078	0.977	0.238	1.299	-0.039	0.982	0.276	1.461
γ_{10}	-0.007	0.954	0.084	0.331	-0.005	0.952	0.105	0.408	-0.007	0.956	0.102	0.404	-0.007	0.959	0.102	0.409	-0.010	0.952	0.101	0.396
ICC = 0.3																				
γ_{01}	0.006	0.962	0.269	1.162	-0.019	0.956	0.275	1.165	-0.020	0.947	0.271	1.137	-0.020	0.976	0.271	1.296	-0.008	0.977	0.290	1.407
γ_{10}	-0.001	0.945	0.086	0.327	0.000	0.948	0.105	0.403	-0.003	0.946	0.102	0.398	-0.003	0.948	0.102	0.403	-0.006	0.951	0.102	0.393

Note:
Abbreviations as listed in Table C.1

TABLE C.3: Simulation results ($N_2 = 30$, $\gamma_{01} = 0.0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.003	0.945	0.188	0.755	0.042	0.939	0.186	0.757	0.014	0.962	0.173	0.766	0.014	0.976	0.173	0.812	0.009	0.963	0.199	0.864
γ_{10}	-0.005	0.953	0.059	0.236	-0.006	0.953	0.073	0.291	-0.007	0.954	0.073	0.290	-0.007	0.954	0.073	0.292	-0.008	0.950	0.072	0.283
ICC = 0.3																				
γ_{01}	-0.004	0.957	0.185	0.758	0.010	0.964	0.185	0.760	-0.003	0.955	0.190	0.769	-0.003	0.964	0.190	0.810	-0.003	0.965	0.192	0.826
γ_{10}	0.001	0.942	0.063	0.237	0.000	0.943	0.077	0.291	-0.003	0.939	0.077	0.291	-0.003	0.940	0.077	0.293	-0.003	0.943	0.076	0.284
MAR																				
ICC = 0.1																				
γ_{01}	0.006	0.941	0.191	0.757	0.039	0.947	0.192	0.761	0.010	0.965	0.177	0.767	0.010	0.972	0.177	0.813	0.003	0.965	0.203	0.870
γ_{10}	0.000	0.944	0.061	0.236	-0.002	0.945	0.075	0.290	-0.003	0.950	0.074	0.288	-0.003	0.951	0.074	0.290	-0.004	0.948	0.073	0.282
ICC = 0.3																				
γ_{01}	0.008	0.969	0.179	0.765	0.027	0.956	0.183	0.768	0.011	0.958	0.186	0.777	0.011	0.968	0.186	0.820	0.010	0.967	0.188	0.836
γ_{10}	0.000	0.944	0.062	0.238	0.000	0.948	0.076	0.293	-0.003	0.954	0.075	0.294	-0.003	0.954	0.075	0.296	-0.003	0.946	0.074	0.286

Note:

Abbreviations as listed in Table C.1

TABLE C.4: Simulation results ($N_2 = 30$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.005	0.950	0.189	0.744	-0.034	0.956	0.187	0.748	-0.052	0.956	0.175	0.767	-0.052	0.966	0.175	0.813	-0.009	0.970	0.204	0.865
γ_{10}	0.000	0.966	0.056	0.231	-0.001	0.967	0.068	0.284	0.001	0.966	0.066	0.283	0.001	0.969	0.066	0.285	-0.003	0.967	0.065	0.277
ICC = 0.3																				
γ_{01}	0.007	0.943	0.188	0.736	-0.008	0.941	0.191	0.740	-0.002	0.938	0.194	0.753	-0.002	0.953	0.194	0.794	0.004	0.951	0.201	0.811
γ_{10}	0.000	0.963	0.057	0.229	0.003	0.956	0.069	0.282	0.001	0.950	0.067	0.281	0.001	0.954	0.067	0.283	0.000	0.954	0.067	0.275
MAR																				
ICC = 0.1																				
γ_{01}	0.005	0.948	0.191	0.744	-0.033	0.944	0.190	0.749	-0.057	0.952	0.174	0.772	-0.057	0.963	0.174	0.819	-0.015	0.958	0.204	0.868
γ_{10}	0.001	0.947	0.060	0.233	0.001	0.946	0.073	0.285	0.003	0.948	0.073	0.285	0.003	0.950	0.073	0.287	0.000	0.946	0.071	0.277
ICC = 0.3																				
γ_{01}	0.002	0.941	0.182	0.737	-0.014	0.947	0.180	0.738	-0.006	0.942	0.185	0.754	-0.006	0.951	0.185	0.795	0.000	0.954	0.191	0.813
γ_{10}	-0.002	0.948	0.059	0.230	-0.001	0.947	0.075	0.284	-0.003	0.950	0.072	0.283	-0.003	0.951	0.072	0.285	-0.004	0.948	0.072	0.277

Note:

Abbreviations as listed in Table C.1

TABLE C.5: Simulation results ($N_2 = 60$, $\gamma_{01} = 0.0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	-0.006	0.955	0.130	0.516	0.027	0.954	0.128	0.517	-0.002	0.971	0.124	0.542	-0.002	0.979	0.124	0.556	-0.010	0.961	0.139	0.575
γ_{10}	0.000	0.944	0.043	0.167	-0.001	0.953	0.053	0.205	-0.002	0.939	0.053	0.205	-0.002	0.940	0.053	0.206	-0.002	0.945	0.052	0.199
ICC = 0.3																				
γ_{01}	-0.002	0.946	0.127	0.518	0.017	0.955	0.127	0.519	0.001	0.949	0.132	0.536	0.001	0.954	0.132	0.549	0.000	0.954	0.133	0.550
γ_{10}	0.000	0.947	0.042	0.167	0.002	0.947	0.052	0.206	0.000	0.948	0.051	0.206	0.000	0.948	0.051	0.206	0.000	0.946	0.051	0.201
MAR																				
ICC = 0.1																				
γ_{01}	0.000	0.951	0.131	0.517	0.037	0.949	0.129	0.520	0.008	0.960	0.126	0.545	0.008	0.965	0.126	0.559	0.001	0.953	0.142	0.582
γ_{10}	-0.001	0.954	0.043	0.167	0.000	0.954	0.052	0.204	-0.001	0.956	0.051	0.205	-0.001	0.957	0.051	0.205	-0.001	0.953	0.050	0.198
ICC = 0.3																				
γ_{01}	0.005	0.948	0.131	0.522	0.021	0.949	0.131	0.522	0.003	0.946	0.137	0.539	0.003	0.949	0.137	0.552	0.003	0.956	0.137	0.556
γ_{10}	0.001	0.954	0.041	0.167	0.000	0.949	0.052	0.205	-0.001	0.947	0.052	0.205	-0.001	0.947	0.052	0.206	-0.001	0.944	0.051	0.201

Note:

Abbreviations as listed in Table C.1

TABLE C.6: Simulation results ($N_2 = 60$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.001	0.946	0.126	0.509	-0.039	0.939	0.124	0.509	-0.044	0.955	0.121	0.541	-0.044	0.960	0.121	0.556	-0.008	0.952	0.136	0.577
γ_{10}	0.001	0.960	0.041	0.164	0.001	0.961	0.050	0.201	0.004	0.954	0.049	0.200	0.004	0.955	0.049	0.200	0.001	0.958	0.049	0.195
ICC = 0.3																				
γ_{01}	-0.002	0.952	0.129	0.501	-0.021	0.947	0.128	0.502	-0.007	0.954	0.132	0.521	-0.007	0.957	0.132	0.534	-0.005	0.962	0.133	0.536
γ_{10}	-0.001	0.954	0.040	0.162	-0.001	0.956	0.049	0.199	-0.002	0.956	0.048	0.199	-0.002	0.956	0.048	0.200	-0.002	0.961	0.048	0.194
MAR																				
ICC = 0.1																				
γ_{01}	0.000	0.954	0.129	0.509	-0.035	0.943	0.128	0.512	-0.044	0.960	0.125	0.544	-0.044	0.962	0.125	0.559	-0.008	0.957	0.144	0.580
γ_{10}	0.002	0.926	0.043	0.163	0.001	0.942	0.051	0.200	0.003	0.941	0.050	0.199	0.003	0.942	0.050	0.200	0.001	0.941	0.050	0.194
ICC = 0.3																				
γ_{01}	-0.005	0.950	0.127	0.503	-0.022	0.956	0.125	0.504	-0.009	0.958	0.130	0.523	-0.009	0.961	0.130	0.536	-0.008	0.962	0.132	0.540
γ_{10}	-0.001	0.953	0.042	0.162	-0.002	0.944	0.051	0.199	-0.003	0.944	0.050	0.199	-0.003	0.945	0.050	0.200	-0.003	0.942	0.050	0.195

Note:

Abbreviations as listed in Table C.1

TABLE C.7: MCSEs of performance measures ($N_2 = 15$, $\gamma_{01} = 0.0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.009	0.007	0.006	0.010	0.009	0.007	0.007	0.010	0.008	0.006	0.006	0.010	0.008	0.004	0.006	0.012	0.009	0.004	0.007	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.009	0.006	0.006	0.010	0.009	0.007	0.006	0.010	0.009	0.007	0.006	0.011	0.009	0.005	0.006	0.012	0.009	0.004	0.006	0.013
γ_{10}	0.003	0.007	0.002	0.001	0.004	0.008	0.002	0.001	0.003	0.008	0.002	0.002	0.003	0.008	0.002	0.002	0.003	0.008	0.002	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.008	0.005	0.006	0.011	0.008	0.004	0.006	0.013	0.009	0.004	0.006	0.016
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.006	0.006	0.013	0.009	0.005	0.007	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001

Note:

All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.8: MCSEs of performance measures ($N_2 = 15$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.009	0.007	0.006	0.010	0.008	0.006	0.006	0.010	0.008	0.006	0.005	0.010	0.008	0.004	0.005	0.013	0.009	0.004	0.006	0.020
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.007	0.006	0.011	0.009	0.006	0.006	0.013	0.009	0.005	0.006	0.016
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.008	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.009	0.006	0.006	0.010	0.009	0.007	0.006	0.010	0.008	0.006	0.005	0.010	0.008	0.005	0.005	0.012	0.009	0.004	0.006	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.006	0.002	0.002	0.003	0.006	0.002	0.002	0.003	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.008	0.006	0.006	0.010	0.009	0.006	0.006	0.010	0.009	0.007	0.006	0.011	0.009	0.005	0.006	0.013	0.009	0.005	0.006	0.014
γ_{10}	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.001	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.002	0.003	0.007	0.002	0.001

Note:

All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.9: MCSEs of performance measures ($N_2 = 30$, $\gamma_{01} = 0.0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.008	0.004	0.004	0.005	0.006	0.004	0.005	0.005	0.005	0.004	0.005	0.006	0.006	0.004	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.004	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.005
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.008	0.002	0.001	0.002	0.008	0.002	0.001	0.002	0.007	0.002	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.005	0.004	0.006	0.006	0.006	0.005	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.006	0.005	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.006	0.006	0.006	0.004	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001

Note:

All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.10: MCSEs of performance measures ($N_2 = 30$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.004	0.006	0.006	0.004	0.005	0.006	0.006	0.004	0.006	0.006	0.005	0.005	0.006
γ_{10}	0.002	0.006	0.001	0.000	0.002	0.006	0.002	0.001	0.002	0.006	0.001	0.001	0.002	0.005	0.001	0.001	0.002	0.006	0.001	0.001
ICC = 0.3																				
γ_{01}	0.006	0.007	0.004	0.004	0.006	0.007	0.004	0.004	0.006	0.008	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.007	0.005	0.005
γ_{10}	0.002	0.006	0.001	0.000	0.002	0.006	0.002	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001
MAR																				
ICC = 0.1																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.004	0.006	0.007	0.004	0.005	0.006	0.006	0.004	0.006	0.006	0.006	0.005	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001
ICC = 0.3																				
γ_{01}	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.005	0.006	0.007	0.004	0.006	0.006	0.007	0.004	0.006
γ_{10}	0.002	0.007	0.001	0.000	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001	0.002	0.007	0.002	0.001

Note:

All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.11: MCSEs of performance measures ($N_2 = 60$, $\gamma_{01} = 0.0$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.005	0.003	0.003	0.004	0.005	0.003	0.003	0.004	0.006	0.003	0.003
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.008	0.001	0.001	0.002	0.008	0.001	0.001	0.002	0.007	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.003	0.004	0.007	0.003	0.002
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000
MAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.003	0.004	0.006	0.003	0.003	0.004	0.007	0.003	0.003
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.006	0.001	0.001	0.002	0.006	0.001	0.001	0.002	0.007	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.003	0.004	0.006	0.003	0.003
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000

Note:

All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

TABLE C.12: MCSEs of performance measures ($N_2 = 60$, $\gamma_{01} = 0.4$)

	CD				LD				MI-R				MI-a				Bayes			
	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW	Bias	Cov	SD	CIW
MCAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.008	0.003	0.002	0.004	0.007	0.003	0.003	0.004	0.006	0.003	0.003	0.004	0.007	0.003	0.003
γ_{10}	0.001	0.006	0.001	0.000	0.002	0.006	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.006	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.003	0.004	0.006	0.003	0.002
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.006	0.001	0.000	0.002	0.006	0.001	0.001	0.002	0.006	0.001	0.001	0.002	0.006	0.001	0.000
MAR																				
ICC = 0.1																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.003	0.004	0.006	0.003	0.003	0.005	0.006	0.003	0.003
γ_{10}	0.001	0.008	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000
ICC = 0.3																				
γ_{01}	0.004	0.007	0.003	0.002	0.004	0.006	0.003	0.002	0.004	0.006	0.003	0.002	0.004	0.006	0.003	0.002	0.004	0.006	0.003	0.002
γ_{10}	0.001	0.007	0.001	0.000	0.002	0.007	0.001	0.000	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.001	0.002	0.007	0.001	0.000

Note:

All entries are Monte Carlo standard errors (MCSEs). Abbreviations as listed in Table C.1.

Appendix D

Default priors and sensitivity analysis

D.1 Default priors in `mdmb`

Default priors in `mdmb` are noninformative. For regression coefficients, a uniform prior on the interval $(-\infty, \infty)$ is approximated by a normal distribution with large variance, i.e., $\mathcal{N}(0, 10^{10})$.

For variance–covariance matrices on level 1 and 2, inverse-Wishart priors are used:

$$\Sigma \sim \mathcal{IW}(\nu, \mathbf{S})$$

where

- Σ denotes a generic covariance matrix (σ at level 1 and τ at level 2),
- ν denotes the degrees of freedom, and
- $\mathbf{S}$ is the scale matrix.

The default in `mdmb` is set to improper inverse-Wishart priors with negative degrees of freedom, which can be written as $\mathcal{IW}(-p - 1, \mathbf{0})$ (Asparouhov & Muthen, 2010), where p denotes the dimension of the covariance matrix. $\mathbf{S} = \mathbf{0}$ corresponds to a null matrix of appropriate dimension.

D.2 Default priors in `jomo`

In `jomo`, inverse-Wishart priors are used for the covariance matrices at level 1 and 2 for imputation. The default in the `mitml` implementation is set to “least informative” proper priors with largest dispersion, i.e., smallest possible degrees of freedom, scaled by the identity matrix (Grund et al., 2015; Quartagno & Carpenter, 2023). For the three variables of the imputation model (X, Y, Z) in this thesis the scale matrix is:

$$\mathbf{S} = \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

TABLE D.1: Bias reduction with alternative priors

N2	Bias.d	rel. Bias.d	Bias.a	rel. Bias.a	Bias red.
MCAR					
15	-0.075	18.7%	-0.048	12.1%	35.5%
30	-0.062	15.4%	-0.033	8.3%	46.2%
60	-0.040	9.9%	-0.014	3.6%	63.7%
MAR					
15	-0.068	17.1%	-0.037	9.1%	46.6%
30	-0.059	14.8%	-0.033	8.2%	44.3%
60	-0.045	11.2%	-0.021	5.3%	52.3%

Note:

d = default priors, a = alternative priors, Bias red. = Bias reduction

D.3 Sensitivity analysis with weakly informative priors in jomo

In this analysis, for the conditions when multiple imputation produced the highest bias, i.e. when $ICC = 0.1$ and $\gamma_{01} = .4$, alternative inverse Wishart priors were specified for the level-2 covariance matrix. This was done by specifying a plausible prior guess for the level-2 covariance matrix. In the simulation study, this guess was based on the true parameter values; in empirical applications, it could be derived from previous research. This guess for the covariance matrix corresponds to the two standardized level-1 variables X and Y ($ICC = .1$) and the standardized level-2 variable W :

$$\tau_{\text{guess}} = \begin{bmatrix} .1 & 0 & 0 \\ 0 & .1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

τ denotes the level-2 covariance matrix of the random effects. The level-2 covariance prior was then defined as

$$\tau \sim \mathcal{IW}(\nu, \mathbf{S}), \quad \mathbf{S} = 3 \tau_{\text{guess}}$$

Only the scale matrix was modified, while the degrees of freedom were kept at their default value, such that the prior remained weakly informative but was centered around more plausible variance values. The level-1 covariance matrix was kept at the default specification.

Bias was substantially reduced across all investigated conditions, with reductions ranging from approximately 35% to over 60% and increasing with larger number of groups (see Table D.1). Coverage rates remained stable, while empirical standard deviations showed a small increase under the alternative prior specification.

Affidavit

"I hereby declare under oath that I have written this thesis independently and have not used any resources, including additional internet sources, other than those specified in the bibliography. I have not previously submitted the thesis for any other examination procedure, and that the written version corresponds to the version on the digital storage medium.

I further declare that I have fully documented all computational/AI-assisted writing tools used in preparing this thesis in the "Overview of Use" section, including product names and how they were used.

Throughout the preparation and writing of this thesis, I have maintained independent authorship and when using computational/AI-supported writing tools, I have exercised direct control and oversight over their application."

Hamburg, May 5, 2026

Signature

Declaration on the Use of Generative Artificial Intelligence

In writing this thesis, I used the following generative artificial intelligence (gAI) systems:

1. ChatGPT
2. DeepL

I further declare that I

- have actively informed myself about the capabilities and limitations of the above-mentioned gAI systems,
- have marked passages taken verbatim from the above-mentioned gAI systems,
- have verified that the content generated using the above-mentioned gAI systems and adopted by me is factually correct,
- am aware that, as the author of this work, I am responsible for the information and statements made therein.

I have used the above-mentioned gAI systems for the purposes described on the next page. The specific system is not listed for each task, as ChatGPT was used across all stages, except for translation and language refinement, where DeepL was additionally used.

Tabular Representation of the Use of AI Systems

Work Step	Description of Use
Programming and Simulation Development	
Debugging	Debugging of R code used in simulation studies
Code understanding	Assistance in understanding and implementing simulation code
Data handling	Support in handling and processing simulation output data
Statistical and Methodological Support	
Conceptual understanding	Assistance in understanding statistical concepts and procedures
Equations	Support in transforming equations
Literature Work	
Literature research	Assistance in identifying relevant literature when needed
Literature understanding	Assistance in understanding complex passages in scientific literature
Data Visualization	
Generating visualizations	Assistance in generating and refining graphical representations of simulation results in R
Equations in diagrams	Assistance with integrating LaTeX equation rendering into Inkscape
Writing and Text Development	
Text style	Improving clarity, structure, and academic style of text passages
Translations	Translating text into English (in combination with DeepL)
Argumentation	Providing suggestions for revision and improvement
Technical Implementation of the Thesis Document	
Markdown/Quarto	Assistance with the use of Quarto and Markdown
Formatting	Support in formatting and adjusting the template (e.g., tables, graphics, layout adjustments, and LaTeX integration)

Hamburg, May 5, 2026

Signature